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1 Convention for adjoint interpolators and overall phase

1.1 Adjoint currents

We the following rule for complex conjugation of a product of Grassmann numbers z_1, z_2

$$(z_1 z_2)^* = z_2^* z_1^*. \quad (1)$$

With this convention, the quark-bilinear $\bar{q}q$ (e.g. part of the mass term) is a real quantity

$$(\bar{q}q)^* = \left(q^\dagger \gamma_t q \right)^* = q^\dagger \gamma_t^\dagger q = \bar{q}q.$$

Adjoint baryon current A baryon interpolator Q is defined by three quark fields q_1, q_2, q_3 and two spin matrices $\Gamma_{1,2}$ and

$$Q_\alpha = \epsilon_{abc} \left(q_1^{aT} \Gamma_1 q_2^b \right) (\Gamma_2 q_3^c)_\alpha = \epsilon_{abc} \left(q_1^a \gamma (\Gamma_1)_{\gamma\delta} q_2^b \right) (\Gamma_2)_{\alpha\alpha'} q_3^c_{\alpha'} \quad (2)$$

Thus for the adjoint baryon interpolating field $\bar{Q}$ we have

$$\begin{aligned} \bar{Q}_\beta &= \left[\epsilon_{a'b'c'} \left(q_1^{a'} \gamma (\Gamma_1)_{\gamma\delta} q_2^{b'} \right) (\Gamma_2)_{\rho\rho'} q_3^{c'}_{\rho'} \right]^* (\gamma_t)_{\rho\beta} \\ &= \epsilon_{a'b'c'} \left(q_2^{b'*} (\Gamma_1^\dagger)_{\delta\gamma} q_1^{a'*} \right) q_3^{c'*} (\Gamma_2^\dagger)_{\rho'\rho} (\gamma_t)_{\rho\beta} \\ &= -\epsilon_{c'b'a'} \bar{q}_3^{c'}_{\beta'} (\gamma_t \Gamma_2^\dagger \gamma_t)_{\beta'\beta} \left(\bar{q}_2^{b'} (\gamma_t)_{\kappa\delta} (\Gamma_1^\dagger)_{\delta\gamma} (\gamma_t)_{\gamma\lambda} \bar{q}_1^{a'}_\lambda \right) \\ &= -\epsilon_{c'a'b'} \left(\bar{q}_3^{c'} \Gamma_2^{t\dagger} \right)_\beta \left(\bar{q}_2^{a'} \Gamma_2^{t\dagger} \bar{q}_1^{b'T} \right), \end{aligned} \quad (3)$$

where

$$\Gamma^{t\dagger} = \gamma_t \Gamma^\dagger \gamma_t = \sigma_\Gamma^{t\dagger} \Gamma$$

and we used the property $\gamma_t^T = \gamma_t$.

Note: We usually multiply another Γ' to q_3 in eq. (3). This will lead to an analogous sign $\sigma_{\Gamma'}^{t\dagger}$.

We used $C^\dagger = C$ in the above table.

Adjoint meson current For a meson interpolating field, this entails

$$(\bar{q}_1 \Gamma q_2)^* = q_2^\dagger \Gamma^\dagger \gamma_t q_1 = \bar{q}_2 \Gamma^{t\dagger} q_1.$$

In particular for the charged pions we have $\pi^+ = \bar{d} i \gamma_5 u$ with $\pi^{+*} = \pi^-$.

Note: I do not include factors of imaginary unit i in the interpolating fields in the contractions. They must be added afterwards as a complex phase to the correlation matrix.

This holds in particular for $C = i \gamma_t \gamma_y$ in the baryon interpolators and $i \gamma_5$ in pion interpolator: in the raw contractions I only consider them as $\gamma_t \gamma_y$ and γ_5 .

1.2 Overall Phase factor for correlation functions

For the Γ structures for the baryon interpolators we use the notation

$$\begin{aligned} i\tilde{C} &= C, \quad \tilde{C} = \gamma_t \gamma_y \\ \Gamma_1 &= C \gamma_1 = i\tilde{C} \gamma_1 = i\tilde{\Gamma}_1 \\ \Gamma_2 &= \tilde{\Gamma}_2 \end{aligned} \tag{4}$$

We write the $\sigma^{t\dagger}$ sign for $\tilde{\Gamma}$, **NOT** Γ . But we have

$$\Gamma_1^{t\dagger} = (-i) \tilde{\Gamma}_1^{t\dagger}$$

Meson spin structure The following table gives the signs $\sigma^{t\dagger}$ for mesonic spin structures.

Meson	$\tilde{\Gamma}$	γ_t	$\gamma_{x,y,z}$	$\mathbb{1}$	γ_5	$\gamma_t \gamma_5$	$\gamma_{x,y,z} \gamma_5$	$\gamma_t \gamma_{x,y,z}$	$\gamma_x \gamma_y, \gamma_x \gamma_z, \gamma_y \gamma_z$
	$\sigma_{\tilde{\Gamma}}^{t\dagger}$	+1	-1	+1	-1	+1	-1	+1	-1

Baryon interpolator (inner) spin structure The sign $\sigma_{\tilde{\Gamma}}^{t\dagger}$ for some relevant spin structures is given below

Nucleon	$\tilde{\Gamma}$	$\tilde{C} \gamma_5$	$\tilde{C}$	$\tilde{C} \gamma_5 \gamma_t$	$\tilde{C} \gamma_t$
	$\sigma_{\tilde{\Gamma}}^{t\dagger}$	-1	+1	-1	-1

Delta	Γ	$\tilde{C} \gamma_{x,z}$	$\tilde{C} \gamma_y$	$\tilde{C} \gamma_{x,z} \gamma_t$	$\tilde{C} \gamma_y \gamma_t$	$\tilde{C} \gamma_{x,z} \gamma_t \gamma_5$	$\tilde{C} \gamma_y \gamma_t \gamma_5$	$\tilde{C} \gamma_{x,z} \gamma_5$	$\tilde{C} \gamma_y \gamma_5$
	$\sigma_{\Gamma}^{t\dagger}$	-1	+1	-1	+1	+1	-1	-1	+1

Summary — overall phase according to convention In the table below, the origin of a sign / phase is given in square brackets **phase [reason]**.

interpolator	at source	at sink
pion	$(-i) \sigma_{\gamma_5}^{t\dagger} [i\gamma_5, \dagger]$	$i [i\gamma_5]$
Nucleon, Delta	$(-1) (-i) \sigma_{\tilde{\Gamma}_1}^{t\dagger} \sigma_{\tilde{\Gamma}_2}^{t\dagger} [\epsilon, C = i\tilde{C}, \dagger]$	$i [C = i\tilde{C}]$

The following table gives the overall phase, according to the above convention, to be multiplied to recombined diagrams.

interpolator	overall phase at source	overall phase at sink
$N \left(\tilde{C}_{\gamma_5}, \mathbb{1} \right)$	$-i$	$+i$
$N \left(\tilde{C}, \gamma_5 \right)$	$-i$	$+i$
$N \left(C \tilde{\gamma}_5 \gamma_t, \mathbb{1} \right)$	$-i$	$+i$
$N \left(\tilde{C}_{\gamma_t}, \gamma_5 \right)$	$+i$	$+i$
$\Delta \left(\tilde{C}_{\gamma_{x,z}}, \mathbb{1} \right)$	$-i$	$+i$
$\Delta \left(\tilde{C}_{\gamma_y}, \mathbb{1} \right)$	$+i$	$+i$
$\Delta \left(\tilde{C}_{\gamma_{x,z} \gamma_t}, \mathbb{1} \right)$	$-i$	$+i$
$\Delta \left(\tilde{C}_{\gamma_y \gamma_t}, \mathbb{1} \right)$	$+i$	$+i$
$\Delta \left(\tilde{C}_{\gamma_{x,z} \gamma_t \gamma_5}, \gamma_5 \right)$	$-i$	$+i$
$\Delta \left(\tilde{C}_{\gamma_y \gamma_t \gamma_5}, \gamma_5 \right)$	$+i$	$+i$
$\Delta \left(\tilde{C}_{\gamma_{x,z} \gamma_5}, \gamma_5 \right)$	$+i$	$+i$
$\Delta \left(\tilde{C}_{\gamma_y \gamma_5}, \gamma_5 \right)$	$-i$	$+i$
$N \left(\tilde{C}_{\gamma_5}, \mathbb{1} \right) \pi$	$+1$	-1
$N \left(\tilde{C}, \gamma_5 \right) \pi$	$+1$	-1
$N \left(\tilde{C}_{\gamma_5 \gamma_t}, \mathbb{1} \right) \pi$	$+1$	-1
$N \left(\tilde{C}_{\gamma_t}, \gamma_5 \right) \pi$	-1	-1

2 Inversions and contractions — practical implementation on the lattice

2.1 Δ^{++} to $\pi^+ N^+$ 2-point function

$$\langle \left[u_\gamma^a(x_f) \left(\tilde{\Gamma}_\Delta \right)_{\gamma\delta} u_\delta^b(x_f) u_\alpha^c(x_f) \right] \left[\bar{u}_\sigma^d(x_{i_2}) (\Gamma_M)_{\sigma\tau} d_\tau^d(x_{i_2}) \right] \left[\bar{d}_\kappa^l(x_{i_1}) \left(\tilde{\Gamma}_N \right)_{\kappa\lambda} \bar{u}_\lambda^m(x_{i_1}) \bar{u}_\beta^n(x_{i_1}) \right] \rangle_f = \quad (5)$$

$$\begin{aligned} & - T(x_f, x_{i_1}) \tilde{\Gamma}_N \left(\tilde{\Gamma}_\Delta U(x_f, x_{i_1}) \right)^t U(x_f, x_{i_1}) \\ & - T(x_f, x_{i_1}) \tilde{\Gamma}_N U(x_f, x_{i_1})^t \tilde{\Gamma}_\Delta U(x_f, x_{i_1}) \\ & - U(x_f, x_{i_1}) \left(\tilde{\Gamma}_\Delta T(x_f, x_{i_1}) \tilde{\Gamma}_N \right)^t U(x_f, x_{i_1}) \\ & - U(x_f, x_{i_1}) \left(T(x_f, x_{i_1}) \tilde{\Gamma}_N \right)^t \tilde{\Gamma}_\Delta U(x_f, x_{i_1}) \\ & - U(x_f, x_{i_1}) \text{Tr} \left(T(x_f, x_{i_1}) \tilde{\Gamma}_N U(x_f, x_{i_1})^t \tilde{\Gamma}_\Delta \right) \\ & - U(x_f, x_{i_1}) \text{Tr} \left(T(x_f, x_{i_1}) \tilde{\Gamma}_N \left(\tilde{\Gamma}_\Delta U(x_f, x_{i_1}) \right)^t \right) \\ & = T_1 + T_2 + T_3 + T_4 + T_5 + T_6 \end{aligned}$$

Eq. (5) defines the triangle diagrams $T_1, \dots, T_6$.

In the following standardized table $\tilde{\Gamma}_N = \tilde{\Gamma}_i$ and $\tilde{\Gamma}_\Delta = \tilde{\Gamma}_f$.

flavor	diag	T 1/2	OP_{i_1}	OP_{f_1}	$OP_{\mathcal{D}}$	IdW	Idtm
$uu\ u - \bar{u}d - \bar{d}\bar{u}\ \bar{u}$	T_1	$\mathcal{T}_1 \left(T, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	N	T	N	T_1	$\tilde{T}_1$
	T_2	$\mathcal{T}_1 \left(T, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	N	N	N	T_2	$\tilde{T}_2$
	T_3	$\mathcal{T}_1 \left(U, T \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	T	T	N	T_3	$\tilde{T}_3$
	T_4	$\mathcal{T}_1 \left(U, T \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	T	N	N	T_3	$\tilde{T}_4$
	T_5	$\mathcal{T}_2 \left(U, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f T \right)$	N	N	N	T_5	$\tilde{T}_5$
	T_6	$\mathcal{T}_2 \left(U, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f T \right)$	N	T	N	T_6	$\tilde{T}_6$

Case of antisymmetric $\tilde{\Gamma}_N$ and symmetric $\tilde{\Gamma}_\Delta$ These are our cases of interest. We then have $T_1 = T_2$, $T_3 = T_4$ and $T_5 = T_6$, so only 3 contributions.

Notation for sequential propagators

$$T(x_f, x_i) = T_{q_1 q_2}(x_f; t, \vec{q}, \Gamma; x_i)_{\alpha\beta}^{ab} = \sum_{\vec{y}} \left(S_{q_1}(x_f; t, \vec{y}) \Gamma e^{i\vec{q}\vec{y}} S_{q_2}(t, \vec{y}; x_i) \right)_{\alpha\beta}^{ab} \quad (6)$$

for the sequential propagator with

- flavors “ q_1 after q_2 ”;
- sequential source timeslice t ;
- sequential source momentum $\vec{q}$;
- sequential source Dirac structure Γ .

We also use the notation

$$T_{fii} = T(x_f; t_i, \vec{q}; x_i) \quad \text{final after initial after initial time} \quad (7)$$

$$T_{ffi} = T(x_f; t_f, \vec{q}; x_i) \quad \text{final after final after initial time} \quad (8)$$

for 1-step sequential propagators.

2.2 Δ^{++} to Δ^{++}

$$\begin{aligned}
& \langle \left[\epsilon_{abc} u_\gamma^a(x_f) \left(\tilde{\Gamma}_f \right)_{\gamma\delta} u_\delta^b(x_f) u_\alpha^c(x_f) \right] \left[\epsilon_{lmn} \bar{u}_\kappa^l(x_i) \left(\tilde{\Gamma}_i \right)_{\kappa\lambda} \bar{u}_\lambda^m(x_i) \bar{u}_\beta^n(x_i) \right] \rangle_f = \quad (9) \\
& \epsilon_{abc} \epsilon_{lmn} \left(\tilde{\Gamma}_f \right)_{\gamma\delta} \left(\tilde{\Gamma}_i \right)_{\kappa\lambda} \left\{ \right. \\
& \quad + U_{\alpha\beta}^{cn} \left(U_{\delta\kappa}^{bl} U_{\gamma\lambda}^{am} - U_{\delta\lambda}^{bm} U_{\gamma\kappa}^{al} \right) \\
& \quad - U_{\alpha\lambda}^{cm} \left(U_{\delta\kappa}^{bl} U_{\gamma\beta}^{an} - U_{\delta\beta}^{bn} U_{\gamma\kappa}^{al} \right) \\
& \quad + U_{\alpha\kappa}^{cl} \left(U_{\delta\lambda}^{bm} U_{\gamma\beta}^{an} - U_{\delta\beta}^{bn} U_{\gamma\lambda}^{am} \right) \\
& \left. \right\} = \\
& - U(x_f, x_i) \tilde{\Gamma}_i \left(\tilde{\Gamma}_f U(x_f, x_i) \right)^t U(x_f, x_i) \\
& - U(x_f, x_i) \tilde{\Gamma}_i U(x_f, x_i)^t \tilde{\Gamma}_f U(x_f, x_i) \\
& - U(x_f, x_i) \left(\tilde{\Gamma}_f U(x_f, x_i) \tilde{\Gamma}_i \right)^t U(x_f, x_i) \\
& - U(x_f, x_i) \left(U(x_f, x_i) \tilde{\Gamma}_i \right)^t \tilde{\Gamma}_f U(x_f, x_i) \\
& - U(x_f, x_i) \text{Tr} \left(\tilde{\Gamma}_f U(x_f, x_i) \tilde{\Gamma}_i U(x_f, x_i)^t \right) \\
& - U(x_f, x_i) \text{Tr} \left(\tilde{\Gamma}_f U(x_f, x_i) \left(U(x_f, x_i) \tilde{\Gamma}_i \right)^t \right) \\
& = D_1 + D_2 + D_3 + D_4 + D_5 + D_6.
\end{aligned}$$

This define the $I = 3/2$ diagrams $D_1, \dots, D_6$.

flavor	diag	T 1/2	OP_{i_1}	OP_{f_1}	$OP_{\mathcal{D}}$	IdW	Idtm
$uu \ u - \bar{u}\bar{u} \ \bar{u}$	D_1	$\mathcal{T}_1 \left(U, N \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	N	T	N	D_1	$\tilde{D}_1$
	D_2	$\mathcal{T}_1 \left(U, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	N	N	N	D_2	$\tilde{D}_2$
	D_3	$\mathcal{T}_1 \left(U, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	T	T	N	D_3	$\tilde{D}_3$
	D_4	$\mathcal{T}_1 \left(U, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	T	N	N	D_3	$\tilde{D}_4$
	D_5	$\mathcal{T}_2 \left(U, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	T	T	N	D_5	$\tilde{D}_5$
	D_6	$\mathcal{T}_2 \left(U, U \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right)$	T	N	N	D_6	$\tilde{D}_6$

Case of symmetric $\tilde{\Gamma}_{i,f}$ for Δ^{++} interpolators Then $\sigma_{\tilde{\Gamma}_f}^T = +1 = \sigma_{\tilde{\Gamma}_i}^T$ and $\sigma_{\tilde{\Gamma}_f}^T \sigma_{\tilde{\Gamma}_i}^T = +1$.

Moreover, $D_1 = D_2 = D_3 = D_4$ and $D_5 = D_6$. (So only two contributions, $4 \times D_1 + 2 \times D_5$.)

2.3 N^+ to N^+

$$\begin{aligned}
& \left\langle \left[\epsilon_{abc} u_\gamma^a(x_f) \left(\tilde{\Gamma}_f \right)_{\gamma\delta} d_\delta^b(x_f) u_\alpha^c(x_f) \right] \left[\epsilon_{lmn} \bar{d}_\kappa^l(x_i) \left(\tilde{\Gamma}_i \right)_{\kappa\lambda} \bar{u}_\lambda^m(x_i) \bar{u}_\beta^n(x_i) \right] \right\rangle_f = \quad (10) \\
& - U(x_f, x_i) \left(\tilde{\Gamma}_f D(x_f, x_i) C \tilde{\Gamma}_i \right)^t U(x_f, x_i) \\
& - U(x_f, x_i) \text{Tr} \left(\left(\tilde{\Gamma}_f D(x_f, x_i) C \tilde{\Gamma}_i \right)^t U(x_f, x_i) \right) \\
& = N_1 + N_2.
\end{aligned}$$

$$= \left(\mathcal{T}_1 \left(U, D \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right) + \mathcal{T}_2 \left(U, D \tilde{\Gamma}_i^T, \tilde{\Gamma}_f U \right) \right) \sigma_{\tilde{\Gamma}_i}^T \sigma_{\tilde{\Gamma}_f}^T \quad (11)$$

This defines the diagrams N_1, N_2 .

flavor	diag	T 1/2	OP_{i_1}	OP_{f_1}	OP_D		IdW	Idtm
$ud\ u - \bar{d}\bar{u}\ \bar{u}$	N_1	$\mathcal{T}_1(U, D\Gamma_i^T, \Gamma_f U)$	T	T	N		N_1	$\tilde{N}_1$
	N_2	$\mathcal{T}_2(U, D\Gamma_i^T, \Gamma_f U)$	T	T	N		N_2	$\tilde{N}_2$

Case of antisymmetric $\tilde{\Gamma}_{f,i}$ for nucleon interpolators Then $\sigma_{\tilde{\Gamma}_f}^T = -1 = \sigma_{\tilde{\Gamma}_i}^T$ and $\sigma_{\tilde{\Gamma}_f}^T \sigma_{\tilde{\Gamma}_i}^T = 1$.

2.4 π^+ to π^+

$$\langle J_{\pi^+}(x_f) J_{\pi^+}(x_i)^\dagger \rangle_f = (-1) \left[(i)(-i) \sigma_{\gamma_5}^{t\dagger} \right] \langle \bar{d} \Gamma_f u(x_f) \bar{u} \Gamma_i d(x_i) \rangle \quad (12)$$

$$= (-1) \left[(i)(-i) \sigma_{\gamma_5}^{t\dagger} \right] m1 \quad (13)$$

The phase in square brackets $\left[(i)(-i) \sigma_{\gamma_5}^{t\dagger} \right]$ is covered by the overall phase per convention.

The (Wick contraction fermion loop) sign (initial (-1)) **must be added** by hand, it is not part of diagram $m1$.

3 Post-poned spin-color reduction and factorization for $I = 3/2$, $I^3 = +3/2$

3.1 Some nomenclature

P_{fii}^{dud} is a second-degree sequential propagator with flavor sequence “down after up after down”. In space-time it runs through the vertices $f_1 \leftarrow f_2 \leftarrow i_2 \leftarrow i_1$.

T_{fii}^{ud} is a sequential propagator with flavor sequence “up after down”. In space-time it runs through the vertices $f_1 \leftarrow i_2 \leftarrow i_1$.

T_{fii}^{du} is a sequential propagator with flavor sequence “down after up”. In space-time it runs through the vertices $f_1 \leftarrow f_2 \leftarrow i_1$.

$\xi^{(fx)}$ is a stochastic timeslice source on timeslice t_{fx} . $\phi^{(fx)}$ is the up-type stochastic propagator originating from a stochastic timeslice source t_{fx} . $\bar{\phi}^{(fx)}$ is the down-type counterpart. The evaluation of those fields at x_{fx} is denoted by subscript fx , e.g. $\phi_{f_1}^{(f_2)}$.

We use γ_5 -hermiticity for the stochastic propagator in the form (we leave out the expectation value on the stochastic samples in the following)

$$\begin{aligned} D(x_{f_1}; x_{f_2}) &= \bar{\phi}_{f_1}^{(f_2)} \xi_{f_2}^{(f_2)} \\ &= \left(\gamma_5 U^\dagger \gamma_5 \right) (x_{f_1}; x_{f_2}) = \gamma_5 U(x_{f_2}; x_{f_1})^\dagger \gamma_5 = \gamma_5 \xi_{f_1}^{(f_1)} \phi_{f_2}^{(f_1)\dagger}. \end{aligned} \quad (14)$$

For the 2-point function diagrams this choice is insignificant, since $t_{f_1} = t_{f_2}$. For the 3-point function diagrams having the stochastic propagator originate on the sink timeslice $t_{f_1} \rightarrow t_f$ will simplify the calculation, since then we have need to close the diagrams at $t_{f_2} \rightarrow t_c$ for all $t_{i_1} = t_{i_2} \leq t_c \leq t_f$.

For the one-end trick in Z -type diagrams we use γ_5 -hermiticity in the form

$$\begin{aligned} T_{f_1 i_2 f_2}^{ud} &= U(x_{f_1}; x_{i_2}) \Gamma_{i_2}(\vec{p}_{i_2}) D(x_{i_2}; x_{f_2}) \\ &= U(x_{f_1}; x_{i_2}) \Gamma_{i_2}(\vec{p}_{i_2}) \gamma_5 U(x_{f_2}; x_{i_2})^\dagger \gamma_5 \\ &= \phi_{f_1}^{(i_2, \alpha)} (\Gamma_{i_2}(\vec{p}_{i_2}) \gamma_5)_{\alpha\beta} \phi_{f_2}^{(i_2, \beta)\dagger} \gamma_5 \end{aligned} \quad (15)$$

We may use $\gamma_5^t = \gamma_5$ where needed.

It is understood, that each diagram is of type spinor propagator, which has open spin indices α, β from sink and source.

3.2 B -diagrams

$$P_{ffii}^{dud} = \bar{\phi}_{f_1}^{(f_2)} \xi_{f_2}^{(f_2)\dagger} \Gamma_{f_2}(\vec{p}_{f_2}) T^{ud}(f_2, i_2, i_1) \quad (16)$$

$$= \gamma_5 \xi_{f_1}^{(f_1)} \phi_{f_2}^{(f_1)\dagger} \gamma_5 \Gamma_{f_2}(\vec{p}_{f_2}) T^{ud}(f_2, i_2, i_1) \quad (17)$$

Diagram B_1

$$(B_1)_{\alpha\beta} = -b_{1\phi}(\vec{p}_{f_1}, \Gamma_{f_1})_{\beta\alpha\delta}^m (\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma}^t b_{1\xi}(\vec{p}_{f_2}, \Gamma_{f_2})_{\gamma}^m$$

$$b_{1\xi;\gamma}^m = \xi(f_2)^\dagger \Gamma_{f_2}(\vec{p}_{f_2}) T_{fii}(f_2, i_1)_{\gamma}^m \quad (18)$$

$$b_{1\phi;\beta\alpha\delta}^m = \epsilon_{mnl} \left[\epsilon_{bca} \phi(f_1)_{\kappa}^b (\Gamma_{f_1})_{\kappa\lambda}^t U(f_1, i_1)_{\lambda,\beta}^{cn} \right] U(f_1, i_1)_{\alpha,\delta}^{al} e^{i\vec{p}_{f_1}} \quad (19)$$

$$B_1 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma}^t \mathcal{V}_3 \left(\phi, \gamma_5 \Gamma_{f_2} T_{fii}^{ud} \right)_{\gamma}^m \times \mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1}^t U, U \right)_{\beta\alpha\delta}^m \quad (20)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

Diagram B_2

$$(B_2)_{\alpha\beta} = -b_{2\phi}(\vec{p}_{f_1}, \Gamma_{f_1})_{\delta\alpha\beta}^m (\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma}^t b_{2\xi}(\vec{p}_{f_2}, \Gamma_{f_2})_{\gamma}^m$$

$$b_{2\xi;\gamma}^m = \xi(f_2)^\dagger \Gamma_{f_2}(\vec{p}_{f_2}) T_{fii}(f_2, i_1)_{\gamma}^m = b_{1\xi;\gamma} \quad (21)$$

$$b_{2\phi;\delta\alpha\beta}^m = \epsilon_{mnl} \left[\epsilon_{bca} \phi(f_1)_{\kappa}^b (\Gamma_{f_1})_{\kappa\lambda}^t U(f_1, i_1)_{\lambda\delta}^{cn} \right] U(f_1, i_1)_{\alpha,\beta}^{al} e^{i\vec{p}_{f_1}}$$

$$= b_{1\phi;\delta\alpha\beta}^m \quad (22)$$

$$B_2 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma}^t \mathcal{V}_3 \left(\phi, \gamma_5 \Gamma_{f_2} T_{fii}^{ud} \right)_{\gamma}^m \times \mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1}^t U, U \right)_{\delta\alpha\beta} \quad (23)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

3.3 W -diagrams

$$T_{ffii}^{du} = \bar{\phi}_{f_1}^{(f_2)} \xi_{f_2}^{(f_2)\dagger} \Gamma_{f_2}(\vec{p}_{f_2}) U(f_2, i_1) \quad (24)$$

$$= \gamma_5 \xi_{f_1}^{(f_1)} \phi_{f_2}^{(f_1)\dagger} \gamma_5 \Gamma_{f_2}(\vec{p}_{f_2}) U(f_2, i_1) . \quad (25)$$

Diagram W_1

$$(W_1)_{\alpha\beta} = -w_{1\phi;\beta\alpha\delta}^m (\Gamma_{i_1})_{\delta\gamma} w_{1\xi;\gamma}^m e^{i\vec{p}_{i_1}} \\ w_{1\xi;\gamma}^m = \xi(f_2)^\dagger \Gamma_{f_2}(\vec{p}_{f_2}) U(f_2, i_1)_{\cdot,\gamma}^m \quad (26)$$

$$w_{1\phi;\beta\alpha\delta}^m = \epsilon_{mnl} \left[\epsilon_{bca} \phi(f_1)_\kappa^b (\Gamma_{f_1})_{\kappa\lambda}^t U(f_1, i_1)_{\lambda\beta}^{cn} \right] T_{fii}(f_1, i_1)_{\alpha\delta}^{al} e^{i\vec{p}_{f_1}} \quad (27)$$

$$W_1 = -(\Gamma_{i_1})_{\delta\gamma} \mathcal{V}_3(\phi, \gamma_5 \Gamma_{f_2} U)_\gamma^m \times \mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1}^t U, T_{fii}^{ud} \right)_{\beta\alpha\delta}^m \quad (28)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

Diagram W_2

$$(W_2)_{\alpha\beta} = -w_{2\phi;\gamma\alpha\delta}^n (\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma} w_{2\xi;\beta}^n \\ w_{2\xi;\beta}^n = \xi(f_2)^\dagger \Gamma_{f_2}(\vec{p}_{f_2}) U(f_2, i_1)_{\cdot,\beta}^n = w_{1\xi;\beta}^n \quad (29)$$

$$w_{2\phi;\gamma\alpha\delta}^n = \epsilon_{nml} \left[\epsilon_{cba} \phi(f_1)_\kappa^c (\Gamma_{f_1})_{\kappa\lambda}^t U(f_1, i_1)_{\lambda\gamma}^{bm} \right] T_{fii}(f_1, i_1)_{\alpha\delta}^{al} e^{i\vec{p}_{f_1}} \\ = w_{1\phi;\gamma\alpha\delta}^n \quad (30)$$

$$W_2 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma} \mathcal{V}_3(\phi, \gamma_5 \Gamma_{f_2} U)_\beta^m \times \mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1}^t U, T_{fii}^{ud} \right)_{\gamma\alpha\delta}^m \quad (31)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

Diagram W_3

$$(W_3)_{\alpha\beta} = -w_{3\phi;\gamma\alpha\delta}^n (\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma}^t w_{3\xi;\beta}^n \\ w_{3\xi;\beta}^n = \xi^\dagger \Gamma_{f_2}(\vec{p}_{f_2}) U(f_2, i_1)_{\cdot,\beta}^n = w_{1\xi;\beta}^n \quad (32)$$

$$w_{3\phi;\gamma\alpha\delta}^n = \epsilon_{nml} \left[\epsilon_{cba} \phi(f_1)_\lambda^c (\Gamma_{f_1})_{\lambda\kappa}^t T_{fii}(f_1, i_1)_{\kappa\gamma}^{bm} \right] U(f_1, i_1)_{\alpha\delta}^{al} e^{i\vec{p}_{f_1}} \quad (33)$$

$$W_3 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma}^t \mathcal{V}_3(\phi, \gamma_5 \Gamma_{f_2} U)_\beta^n \times \mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1}^t T_{fii}^{ud}, U \right)_{\gamma\alpha\delta}^n \quad (34)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

Diagram W_4

$$\begin{aligned} (W_4)_{\alpha\beta} &= -w_{4\phi; \gamma\alpha\beta}^n (\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma}^t w_{4\xi; \delta}^n \\ w_{4\xi; \delta}^n &= \xi^\dagger \Gamma_{f_2}(\vec{p}_{f_2}) U(f_2, i_1)_{\cdot, \delta}^n = w_{1\xi; \delta}^n \end{aligned} \quad (35)$$

$$\begin{aligned} w_{4\phi; \gamma\alpha\beta}^n &= \epsilon_{nml} \left[\epsilon_{cba} \phi(f_1)_\lambda^c (\Gamma_{f_1})_{\lambda\kappa}^t T_{fii}(f_1, i_1)_{\kappa, \gamma}^{bm} \right] U(f_1, i_1)_{\alpha\beta}^{al} e^{i\vec{p}_{f_1}} \\ &= w_{3\phi; \gamma\alpha\beta}^n \end{aligned} \quad (36)$$

$$W_4 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\delta\gamma}^t \mathcal{V}_3(\phi, \gamma_5 \Gamma_{f_2} U)_\delta^m \times \mathcal{V}_2(\gamma_5 \xi, \Gamma_{f_1}^t T_{fii}^{ud}, U)_{\gamma\alpha\beta} \quad (37)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

Exchanging $\bar{\phi} \rightarrow \phi$

Just keep this in mind:

In all the W diagrams we exchange $\bar{\phi} \xi^\dagger \rightarrow (\gamma_5 \xi) (\gamma_5 \phi)^\dagger$ and thereby achieve the separation of sequential propagators T_{fii} and stochastic propagators $\phi/\bar{\phi}$ into separate diagrams. Thus the contractions can be split accordingly into

- one part using only forward and sequential propagators and stochastic sources;
- one part using only forward and stochastic propagators.

In the $SU(2)$ symmetric case, this comes at no additional cost or complexity, since $\phi = \bar{\phi}$. We just change $\Gamma_{f_2} \rightarrow \gamma_5 \Gamma_{f_2}$ and $\Gamma_{f_1} \rightarrow \gamma_5 \Gamma_{f_1} = \gamma_5 \Gamma_{f_1}$.

3.4 Z -diagrams

$$\begin{aligned} P_{fifi}^{udu}(x_{f_1}; t_{i_2}, \vec{p}_{i_2}, \Gamma_{i_2}; t_{f_2}, \vec{p}_{f_2}, \Gamma_{f_2}; x_{i_1}) \\ = \phi^{(t_{i_2}, 0, \kappa)}(x_{f_1}) (\Gamma_{i_2} \gamma_5)_{\kappa\lambda} \phi^{(t_{i_2}, -\vec{p}_{i_2}, \lambda)}(x_{f_2})^\dagger \gamma_5 \Gamma_{f_2}(\vec{p}_{f_2}) U(x_{f_2}; x_{i_1})_{\cdot, \gamma}^l. \end{aligned} \quad (38)$$

Note. We partially omit the $(\Gamma_{i_2}(\vec{p}_{i_2}) \gamma_5)$ in the following.

Diagram Z_1

$$\begin{aligned} (Z_1)_{\alpha\beta} &= -z_{1\xi; \gamma}^l (\Gamma_{i_1}(\vec{p}_{i_1}))_{\gamma\delta}^t z_{1\phi; \alpha\delta\beta}^l \\ z_{1\xi; \gamma}^l &= \phi^{(t_{i_2}, -\vec{p}_{i_2}, \lambda)}(x_{f_2})^\dagger \gamma_5 \Gamma_{f_2}(\vec{p}_{f_2}) U(x_{f_2}; x_{i_1})_{\cdot, \gamma}^l \end{aligned} \quad (39)$$

$$z_{1\phi; \alpha\delta\beta}^l = \phi^{(t_{i_2}, 0, \kappa)}(x_{f_1})_\alpha^a \left[\epsilon_{abc} \epsilon_{lmn} D(x_{f_1}; x_{i_1})_{\delta, \kappa}^{bm} (\Gamma_{f_1})_{\kappa\lambda}^t U(x_{f_1}; x_{i_1})_{\lambda\beta}^{cn} \right] e^{i\vec{p}_{f_1}} \quad (40)$$

$$Z_1 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\gamma\delta}^t (\Gamma_{i_2} \gamma_5)_{\kappa\lambda} \mathcal{V}_3 \left(\phi^{(t_{i_2}, -\vec{p}_{i_2}, \lambda)}, \gamma_5 \Gamma_{f_2} U \right)_\gamma^l \times \mathcal{V}_4 \left(\phi^{(t_{i_2}, 0, \kappa)}, D, \Gamma_{f_1}^t U \right)_{\alpha\delta\beta}^l \quad (41)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

Diagram Z_2

$$(Z_2)_{\alpha\beta} = -z_{2\phi; \alpha\delta\gamma}^l (\Gamma_{i_1}(\vec{p}_{i_1}))_{\gamma\delta}^t z_{2\xi; \beta}^l \quad (42)$$

$$z_{2\xi; \beta}^l = z_{1\xi; \beta}^l \quad (43)$$

$$z_{2\phi; \alpha\delta\gamma}^l = z_{1\phi; \alpha\delta\gamma}^l$$

$$Z_2 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\gamma\delta}^t (\Gamma_{i_2} \gamma_5)_{\kappa\lambda} \mathcal{V}_3 \left(\phi^{(t_{i_2}, -\vec{p}_{i_2}, \lambda)}, \gamma_5 \Gamma_{f_2} U \right)_\beta^l \times \mathcal{V}_4 \left(\phi^{(t_{i_2}, 0, \kappa)}, D, \Gamma_{f_1}^t U \right)_{\alpha\delta\gamma}^l \quad (44)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

Diagram Z_3

$$(Z_3)_{\alpha\beta} = -z_{3\phi; \delta\alpha\gamma}^n (\Gamma_{i_1}(\vec{p}_{i_1}))_{\gamma\delta}^t z_{3\xi; \beta}^n \quad (45)$$

$$z_{3\xi; \beta}^n = z_{1\xi; \beta}^n$$

$$z_{3\phi; \delta\alpha\gamma}^n = \epsilon_{nml} \left[\epsilon_{cba} \phi^{(t_{i_2}, 0, \kappa)}(x_{f_1})_\kappa^c (\Gamma_{f_1})_{\kappa\lambda} D(x_{f_1}; x_{i_1})_{\lambda\delta}^{bm} \right] U(x_{f_1}; x_{i_1})_{\alpha\gamma}^{al} e^{i\vec{p}_{f_1}} \quad (46)$$

$$Z_3 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\gamma\delta}^t (\Gamma_{i_2} \gamma_5)_{\kappa\lambda} \mathcal{V}_3 \left(\phi^{(t_{i_2}, -\vec{p}_{i_2}, \lambda)}, \gamma_5 \Gamma_{f_2} U \right)_\beta^l \times \mathcal{V}_2 \left(\phi^{(t_{i_2}, 0, \kappa)}, \Gamma_{f_1} D, U \right)_{\delta\alpha\gamma} \quad (47)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

Diagram Z_4

$$(Z_4)_{\alpha\beta} = -z_{4\xi; \gamma}^n (\Gamma_{i_1}(\vec{p}_{i_1}))_{\gamma\delta}^t z_{4\phi; \delta\alpha\beta}^n \quad (48)$$

$$z_{4\xi; \gamma}^n = z_{1\xi; \gamma}^n$$

$$z_{4\phi; \delta\alpha\beta}^n = \epsilon_{nml} \left[\epsilon_{cba} \phi^{(t_{i_2}, 0, \kappa)}(x_{f_1})_\kappa^c (\Gamma_{f_1})_{\kappa\lambda} D(x_{f_1}; x_{i_1})_{\lambda\delta}^{bm} \right] U(x_{f_1}; x_{i_1})_{\alpha\beta}^{al} e^{i\vec{p}_{f_1}} \quad (49)$$

$$= z_{3\phi; \delta\alpha\beta}^n$$

$$Z_4 = -(\Gamma_{i_1}(\vec{p}_{i_1}))_{\gamma\delta}^t (\Gamma_{i_2} \gamma_5)_{\kappa\lambda} \mathcal{V}_3 \left(\phi^{(t_{i_2}, -\vec{p}_{i_2}, \lambda)}, \gamma_5 \Gamma_{f_2} U \right)_\gamma^l \times \mathcal{V}_2 \left(\phi^{(t_{i_2}, 0, \kappa)}, \Gamma_{f_1} D, U \right)_{\delta\alpha\beta} \quad (50)$$

Γ_{i_1} and $\vec{p}_{i_1}$ remain open.

3.5 Triangle diagrams with $\Delta - \pi N$ with πN at sink

$$T_1 = -(\Gamma_i(\vec{p}_i))_{\gamma\delta} \mathcal{V}_2(\bar{\phi}, \Gamma_{f_1}^t(\vec{p}_{f_1})U, U)_{\delta\alpha\gamma}^n \mathcal{V}_3(\xi, \Gamma_{f_2}U)_{\beta}^n \quad (51)$$

$$T_2 = -(\Gamma_i(\vec{p}_i)^t)_{\gamma\delta} \mathcal{V}_2(\bar{\phi}, \Gamma_{f_1}^t(\vec{p}_{f_1})U, U)_{\delta\alpha\gamma}^n \mathcal{V}_3(\xi, \Gamma_{f_2}U)_{\beta}^n \quad (52)$$

$$T_3 = -(\Gamma_i(\vec{p}_i))_{\gamma\delta} \mathcal{V}_2(\bar{\phi}, \Gamma_{f_1}^t(\vec{p}_{f_1})U, U)_{\beta\alpha\gamma}^n \mathcal{V}_3(\xi, \Gamma_{f_2}U)_{\delta}^n \quad (53)$$

$$T_4 = -(\Gamma_i(\vec{p}_i)^t)_{\gamma\delta} \mathcal{V}_2(\bar{\phi}, \Gamma_{f_1}^t(\vec{p}_{f_1})U, U)_{\beta\alpha\gamma}^n \mathcal{V}_3(\xi, \Gamma_{f_2}U)_{\delta}^n \quad (54)$$

$$T_5 = -(\Gamma_i(\vec{p}_i))_{\gamma\delta} \mathcal{V}_2(\bar{\phi}, \Gamma_{f_1}^t(\vec{p}_{f_1})U, U)_{\gamma\alpha\beta}^n \mathcal{V}_3(\xi, \Gamma_{f_2}(\vec{p}_{f_2})U)_{\delta}^n \quad (55)$$

$$T_6 = -(\Gamma_i(\vec{p}_i)^t)_{\gamma\delta} \mathcal{V}_2(\bar{\phi}, \Gamma_{f_1}^t(\vec{p}_{f_1})U, U)_{\gamma\alpha\beta}^n \mathcal{V}_3(\xi, \Gamma_{f_2}(\vec{p}_{f_2})U)_{\delta}^n \quad (56)$$

$\Gamma_i(\vec{p}_i)$ remain open.

Note: We use symmetric Γ_i due to isospin $I = 3/2$, $I_3 = +3/2$ for the Δ^{++} . Thus $\Gamma_i = \Gamma_i^t$ and we have effectively 3 contributions.

Note: $\phi = \bar{\phi}$ with isospin-symmetric action.

Note: $\Gamma_{f_1}^t$ appears in all 6 contributions. Mind minus sign for antisymmetric Γ_{f_1} .

3.6 Conclusion I — required reduction operations

For the B , W , Z -type diagrams we need

reduction type	reduction op	diagrams
$\mathcal{V}_3$	$\mathcal{V}_3 \left(\phi, \gamma_5 \Gamma_{f_2} T_{fii}^{ud} \right)$	$B_1^{2pt}, B_2^{2pt},$
	$\mathcal{V}_3 \left(\phi, \gamma_5 \Gamma_{f_2} U \right)$	$W_1^{2pt}, W_2^{2pt}, W_3^{2pt}, W_4^{2pt}$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} U, U \right)$	$B_1^{2pt}, B_2^{2pt},$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} U, T_{fii}^{ud} \right)$	$W_1^{2pt}, W_2^{2pt},$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} T_{fii}^{ud}, U \right)$	$W_3^{2pt}, W_4^{2pt},$

reduction type	reduction op	diagrams
$\mathcal{V}_3$	$\mathcal{V}_3 \left(\phi^{(i_2, \lambda)}, \gamma_5 \Gamma_{f_2} U \right)$	$Z_1^{2pt}, Z_2^{2pt}, Z_3^{2pt}, Z_4^{2pt}$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\phi^{(i_2, \kappa)}, \Gamma_{f_1} D, U \right)$	Z_3^{2pt}, Z_4^{2pt}
$\mathcal{V}_4$	$\mathcal{V}_4 \left(\phi^{(i_2, \kappa)}, \Gamma_{f_1} D, U \right)$	Z_1^{2pt}, Z_2^{2pt}

3.7 Conclusion II - minimal diagram construction for 2-point functions

flavor	diag	F 2/4	F 3	OP_{i_1}	OP_{f_1}	OP_D	π	IdW	Idtm
$u\bar{d}u - \bar{d}u - \bar{u}d - \bar{d}\bar{u}\bar{u}$	$B_1^{2\text{pt}}$	$\mathcal{V}_2(\phi_u, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_{f_2} T_{f_{ii}}^{ud})$	T	T	T	3412	$B_1^{2\text{pt}}$	$\tilde{B}_1^{2\text{pt}}$
	$B_2^{2\text{pt}}$	$\mathcal{V}_2(\phi_u, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_{f_2} T_{f_{ii}}^{ud})$	T	T	N	1342	$B_2^{2\text{pt}}$	$\tilde{B}_1^{2\text{pt}}$
	$W_1^{2\text{pt}}$	$\mathcal{V}_2(\xi, \gamma_5 \Gamma_{f_1}^T U, T_{f_{ii}}^{ud})$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_{f_2} U)$	N	T	T	3412	$W_1^{2\text{pt}}$	$\tilde{W}_1^{2\text{pt}}$
	$W_2^{2\text{pt}}$	$\mathcal{V}_2(\xi, \gamma_5 \Gamma_{f_1}^T U, T_{f_{ii}}^{ud} U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_{f_2} U)$	T	T	N	1324	$W_2^{2\text{pt}}$	$\tilde{W}_2^{2\text{pt}}$
	$W_3^{2\text{pt}}$	$\mathcal{V}_2(\xi, \gamma_5 \Gamma_{f_1}^T T_{f_{ii}}^{ud}, U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_{f_2} U)$	N	T	N	1324	$W_3^{2\text{pt}}$	$\tilde{W}_3^{2\text{pt}}$
	$W_4^{2\text{pt}}$	$\mathcal{V}_2(\xi, \gamma_5 \Gamma_{f_1}^T T_{f_{ii}}^{ud}, U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_{f_2} U)$	N	T	N	1342	$W_4^{2\text{pt}}$	$\tilde{W}_4^{2\text{pt}}$
	$Z_1^{2\text{pt}}$	$\mathcal{V}_4(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_{f_2} U)$	N	N	N	3142	$Z_1^{2\text{pt}}$	$\tilde{Z}_1^{2\text{pt}}$
	$Z_2^{2\text{pt}}$	$\mathcal{V}_4(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_{f_2} U)$	N	N	N	3124	$Z_2^{2\text{pt}}$	$\tilde{Z}_2^{2\text{pt}}$
	$Z_3^{2\text{pt}}$	$\mathcal{V}_2(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_{f_2} U)$	N	N	N	1324	$Z_3^{2\text{pt}}$	$\tilde{Z}_3^{2\text{pt}}$
	$Z_4^{2\text{pt}}$	$\mathcal{V}_2(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_{f_2} U)$	N	N	N	1342	$Z_4^{2\text{pt}}$	$\tilde{Z}_4^{2\text{pt}}$

flavor	diag	F 2/4	F 3	OP_{i_1}	OP_{f_1}	$OP_{\mathcal{D}}$	π	IdW	Idtm
$udu - \bar{d}u - \bar{u}\bar{u}\bar{u}$	T_1	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_{f_2} U)$	T	T	N	1324	T_1	$\tilde{T}_1$
	T_2	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_{f_2} U)$	N	T	N	1324	T_2	$\tilde{T}_2$
	T_3	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_{f_2} U)$	N	T	T	3412	T_3	$\tilde{T}_3$
	T_4	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_{f_2} U)$	T	T	T	3412	T_4	$\tilde{T}_4$
	T_5	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_{f_2} U)$	N	T	N	1342	T_5	$\tilde{T}_5$
	T_6	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_{f_2} U)$	T	T	N	1342	T_6	$\tilde{T}_6$

3.8 Phase conventions

The sign and phase for each diagram follows from the overall phase (by convention) and the signs from the recombination table.

Direct diagrams The phase for the direct diagram $\langle N N \rangle_f \times \langle \pi \pi \rangle$ follows from the phases of the individual factors. Not special treatment will be needed.

3.9 Reduction operations

3.9.1 $\mathcal{V}_3$ -type: fermion - propagator scalar products

These are of the form

$$V_3(x, y)_\beta^b = \mathcal{V}_3(\xi, S) = \sum_{a, \alpha} \sum_{\vec{x}} \xi(t_x, \vec{x})_\alpha^{a*} \Gamma e^{i\vec{p}\vec{x}} S(t_x, \vec{x}; t_y, \vec{y})_{\alpha\beta}^{ab} \quad (57)$$

This is the same we use for the meson-meson contractions.

3.9.2 $\mathcal{V}_4$ -type: propagator - propagator ϵ products

$$A(x, y)_{\alpha_{i_1}, \alpha_{i_2}}^{al} = \sum_{b, c} \sum_{m, n} \sum_{\alpha_{i_3}, \alpha_{i_4}} \epsilon_{abc} \epsilon_{lmn} S(x, y)_{\alpha_1 \alpha_2}^{bm} S(x, y)_{\alpha_3 \alpha_4}^{cn} \delta_{\alpha_{i_3}, \alpha_{i_4}} \quad (58)$$

with $i_1, i_2, i_3, i_4 \in \{1, 2, 3, 4\}$ and $i_k \neq i_l \quad \forall k \neq l$ and $i_1 < i_2, i_3 < i_4$.

This is *qcd.quarkContractV4aux* or *fp-eps-eps-contractAB-fp*. With the special choice of $A = 1, B = 3$ we get $\mathcal{V}_4$.

$$V_4(x, y)_{\alpha\beta\gamma}^l = \mathcal{V}_4(\phi, S, S) = \sum_a \phi(x)_\alpha^a \text{contractV4aux}(S(x, y), S(x, y))_{\beta\gamma}^{al} . \quad (59)$$

Difference to *qcd.quarkContract13* is the switch of color indices from source side to sink-side and vice versa in *contract13*, i.e.

$$\text{contractV4aux}(S(x, y), S(x, y))_{\beta\gamma}^{al} = \text{contract13}(S(x, y), S(x, y))_{\beta\gamma}^{la} .$$

Symmetries of $\mathcal{V}_4$ We have

$$\begin{aligned} \mathcal{V}_4(\phi, X, Y)_{\alpha_1 \alpha_2 \alpha_3} &= \phi_{\alpha_1}^a \epsilon_{abc} \epsilon_{lmn} X_{\kappa \alpha_2}^{bm} Y_{\kappa \alpha_3}^{cn} \\ &= \phi_{\alpha_1}^a \epsilon_{acb} \epsilon_{lmn} Y_{\kappa \alpha_3}^{cn} X_{\kappa \alpha_2}^{bm} \\ &= \mathcal{V}_4(\phi, Y, X)_{\alpha_1 \alpha_3 \alpha_2} . \end{aligned} \quad (60)$$

Also

$$\begin{aligned} \mathcal{V}_4(\phi, \Gamma X, Y)_{\alpha_1 \alpha_2 \alpha_3} &= \phi_{\alpha_1}^a \epsilon_{abc} \epsilon_{lmn} \Gamma_{\kappa \lambda} X_{\lambda \alpha_2}^{bm} Y_{\kappa \alpha_3}^{cn} \\ &= \phi_{\alpha_1}^a \epsilon_{abc} \epsilon_{lmn} X_{\lambda \alpha_2}^{bm} \Gamma_{\lambda \kappa}^t Y_{\kappa \alpha_3}^{cn} \\ &= \mathcal{V}_4(\phi, X, \Gamma^t Y)_{\alpha_1 \alpha_2 \alpha_3} . \end{aligned} \quad (61)$$

3.9.3 $\mathcal{V}_2$ -type: fermion - propagator ϵ product

This reduction will be useful for a fermion ϕ and a point-to-all propagator S .

$$V_1(x, y)_{\alpha_2}^{am} = \mathcal{V}_1(\phi, S) = \sum_{\alpha_1} \sum_{c, b} \epsilon_{cba} \phi(x)_{\alpha_1}^c S(x, y)_{\alpha_1 \alpha_2}^{bm}. \quad (62)$$

The V_1 s is a mixed type with 2 color indices and 1 Dirac index. (Can be done per spin-color component (α_2, m) if need be).

The final reduction step will then be of the form

$$V_2(x, y)_{\alpha_1 \alpha_2 \alpha_3}^n = \mathcal{V}_2(V_1, S) = \sum_a \sum_{l, m} \epsilon_{nml} V_1(x, y)_{\alpha_1}^{am} S(x, y)_{\alpha_2 \alpha_3}^{al}, \quad (63)$$

which gives again a mixed type with 1 color index and 3 Dirac indices.

V_2 will be Fourier transformed and stored to disk.

Symmetries of $\mathcal{V}_2$ We have

$$\mathcal{V}_2(\phi, \Gamma X, Y) = \mathcal{V}_2(\Gamma^\dagger \phi, X, Y) \quad (64)$$

3.9.4 Complete 3-propagator products

$$\mathcal{T}_1(A, B, C)_{\alpha\beta} = -A_{\alpha\sigma_1} B_{\sigma_1\sigma_2}^t C_{\sigma_2\beta} \quad (65)$$

$$\begin{aligned} \mathcal{T}_2(A, B, C)_{\alpha\beta} &= -A_{\alpha\beta} B_{\sigma_1\sigma_2}^t C_{\sigma_2\sigma_1} \\ &= \mathcal{T}_2(A, C, B)_{\alpha\beta} \end{aligned} \quad (66)$$

Feeding $\Gamma_{1,2}$ into reduction The standard contraction we calculate is: for given propagators A, B, C and Γ -matrices Γ_1, Γ_2

$$-A \Gamma_1 B^t \Gamma_2 C = \mathcal{T}_1(A, B \Gamma_1^t, \Gamma_2 C) \quad (67)$$

$$-A \text{tr}(\Gamma_1 B^t \Gamma_2 C) = \mathcal{T}_2(A, B \Gamma_1^t, \Gamma_2 C) \quad (68)$$

Symmetries of $\mathcal{T}_1$

$$\mathcal{T}_1(A \Gamma, B, C)_{\alpha\beta} = - (A \Gamma B^t C)_{\alpha\beta} = - \left(A (B \Gamma^t)^t C \right)_{\alpha\beta} = \mathcal{T}_1(A, B \Gamma^t, C)_{\alpha\beta} \quad (69)$$

$$\mathcal{T}_1(A, \Gamma B, C)_{\alpha\beta} = - (A B^t \Gamma^t C)_{\alpha\beta} = \mathcal{T}_1(A, B, \Gamma^t C)_{\alpha\beta} \quad (70)$$

Symmetries of $\mathcal{T}_2$

$$\mathcal{T}_2(A, B \Gamma, C) = \mathcal{T}_2(A, B, C \Gamma^t) \quad (71)$$

$$\mathcal{T}_2(A, \Gamma B, C) = \mathcal{T}_2(A, B, \Gamma^t C) \quad (72)$$

3.9.5 Implementation

We will thus need the functions $\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3$ in eqs. (62), (63) above.

In *QLUA* The reduction operations are implemented as functions called by

$\mathcal{V}_1$: ColorVector[36] $v1 = \text{qcd.quarkContractV1}(\text{DiracFermion } F, \text{DiracPropagator } P)$
indexing $v1(a, b, \alpha) = v1[4 \cdot (3 \cdot a + b) + \alpha]$

$\mathcal{V}_2$: ColorVector[192] $v2 = \text{qcd.quarkContractV2}(\text{ColorVector[36] } v1, \text{DiracPropagator } P)$
indexing $v2(\alpha, \beta, \gamma, a) = v2[3 \cdot (4 \cdot (4 \cdot \alpha + \beta) + \gamma) + a]$

$\mathcal{V}_3$ ColorVector[12] $v3 = \text{qcd.quarkContractV3}(\text{DiracFermion } F, \text{DiracPropagator } P)$
indexing $v3(\alpha, a) = v3[3 \cdot \alpha + a]$

$\mathcal{V}_4$ auxilliary DiracPropagator $Q = \text{qcd.quarkContractV4aux}(\text{DiracPropagator } P_1, \text{DiracPropagator } P_2)$
indexing of elements as for DiracPropagator

$\mathcal{V}_4$ ColorVector[192] $v4 = \text{qcd.quarkContractV4}(\text{DiracFermion } F, \text{DiracPropagator } P)$
 $\text{qcd.quarkContractV4}$ uses output of $\text{qcd.quarkContractV4aux}$ as second argument
indexing $v4(\alpha, \beta, \gamma, a) = v2[3 \cdot (4 \cdot (4 \cdot \alpha + \beta) + \gamma) + a]$
(same running of indices as in $v2$)

$\mathcal{T}_1$ ColorVector[16] $t1 = \text{qcd.quarkContractT1}(\text{DiracPropagator } P_1, \text{DiracPropagator } P_2)$
 $\text{qcd.quarkContractV4}$ uses output of $\text{qcd.quarkContractV4aux}$ as second argument
indexing $t1(\alpha, \beta) = t1[4 \cdot \alpha + \beta]$

$\mathcal{T}_2$ ColorVector[16] $t2 = \text{qcd.quarkContractT2}(\text{DiracPropagator } P_1, \text{DiracPropagator } P_2)$
 $\text{qcd.quarkContractV4}$ uses output of $\text{qcd.quarkContractV4aux}$ as second argument
indexing $t2(\alpha, \beta) = t2[4 \cdot \alpha + \beta]$

3.9.6 Summary

Below we list the factors we calculate.

$B-$, W -type diagrams

$$\mathcal{V}_3 : \phi \times \left\{ \begin{array}{c} \gamma_5 \Gamma_c \\ \gamma_5 \Gamma_{f_2} \end{array} \right\} \times \left\{ \begin{array}{c} U \\ T \end{array} \right\} \quad (73)$$

$$\mathcal{V}_2 : \gamma_5 \xi \times \Gamma_{f_1} \left\{ \begin{array}{cc} U & \times U \\ U & \times T \\ T & \times U \end{array} \right\} \quad (74)$$

$$\mathcal{V}_4 : \gamma_5 \xi \times \Gamma_{f_1} \left\{ \begin{array}{cc} U & \times U \\ U & \times T \\ T & \times U \end{array} \right\} \quad (75)$$

Z -type diagrams

$$\mathcal{V}_3 : \phi \times \left\{ \begin{array}{c} \gamma_5 \Gamma_c \\ \gamma_5 \Gamma_{f_2} \end{array} \right\} \times U \quad (76)$$

$$\mathcal{V}_2 : \gamma_5 \phi \times \Gamma_{f_1} U \times U \quad (77)$$

$$\mathcal{V}_4 : \gamma_5 \phi \times \Gamma_{f_1} U \times U \quad (78)$$

We assign non-zero momentum $-\vec{p}_{i_2}$ to ϕ in eq. (76)¹ and zero momentum to ϕ in eqs. (77) and (78).

N , D , S and $D1cc$ -type diagrams

$$\mathcal{T}_1 : \left\{ \begin{array}{ccc} U & \times \Gamma_{f_1} U \Gamma_{i_1} & \times U \\ T & \times \Gamma_{f_1} U \Gamma_{i_1} & \times U \\ U & \times \Gamma_{f_1} T \Gamma_{i_1} & \times U \\ U & \times \Gamma_{f_1} U \Gamma_{i_1} & \times T \end{array} \right\} \quad (79)$$

$$\mathcal{T}_2 : \left\{ \begin{array}{ccc} U & \times \Gamma_{f_1} U \Gamma_{i_1} & \times U \\ T & \times \Gamma_{f_1} U \Gamma_{i_1} & \times U \\ U & \times \Gamma_{f_1} T \Gamma_{i_1} & \times U \\ U & \times \Gamma_{f_1} U \Gamma_{i_1} & \times T \end{array} \right\} \quad (80)$$

$$(81)$$

Note, that in the light quark sector with $SU(2)$ flavor symmetry, this is all we actually can calculate.

- Γ_{f_1} are from $\{C, C\gamma_5, C\gamma_0, C\gamma_5\gamma_0\}$ (nucleon)
- for $\mathcal{T}_{1,2}$ $\Gamma_{f_1}, \Gamma_{i_1}$ are in addition $\{C\vec{\gamma}\}$ (Delta)
- Γ_{f_2} is from $\{\mathbb{1}, \gamma_5\}$ (scalar, pseudo-scalar)
- Γ_c is from $\{\vec{\gamma}, \vec{\gamma}\gamma_5\}$ (vector, axial-vector)

¹Note, that ϕ enters eq. (76) as $\phi^\dagger$; hence we assign $-\vec{p}_{i_2}$ here to obtain a complete diagram with $+\vec{p}_{i_2}$ as is our convention.

4 Contractions for $I = 3/2$, $I_3 = +1/2$ 3-point functions

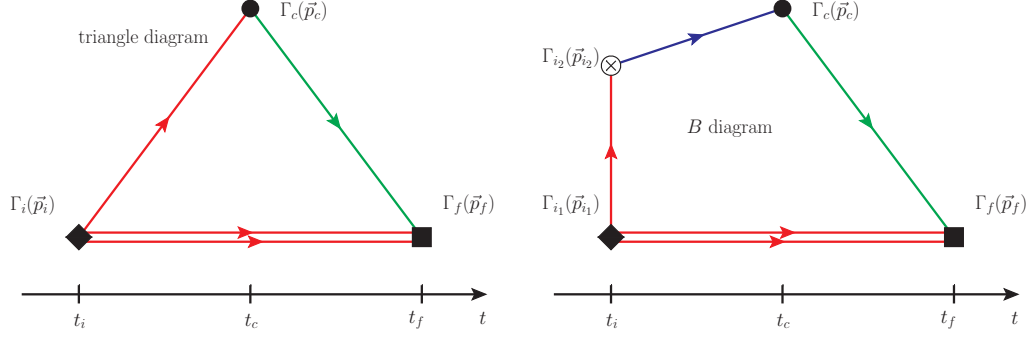


Figure 1: Triangle and B diagram types with nomenclature used in the contraction formulas.

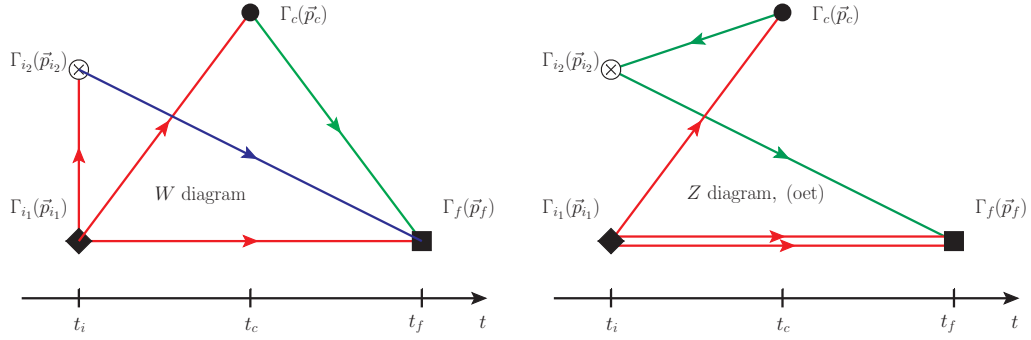


Figure 2: W and Z diagram types with nomenclature used in the contraction formulas.

4.1 Single-hadron interpolating fields

$$J_{\Delta, \Gamma} = \frac{1}{\sqrt{3}} \epsilon_{abc} \left[2 \left(u^{at} \Gamma d^b \right) u^c + \left(u^{at} \Gamma u^b \right) d^c \right] \quad (82)$$

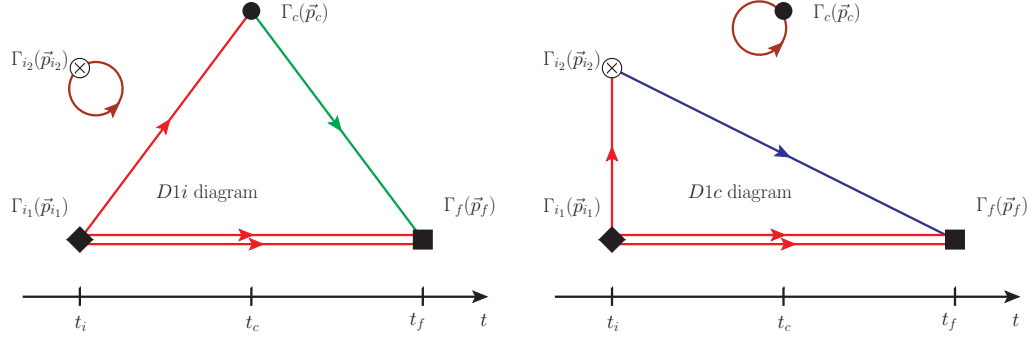


Figure 3: $D1i$ and $D1c$ diagram types with nomenclature used in the contraction formulas.

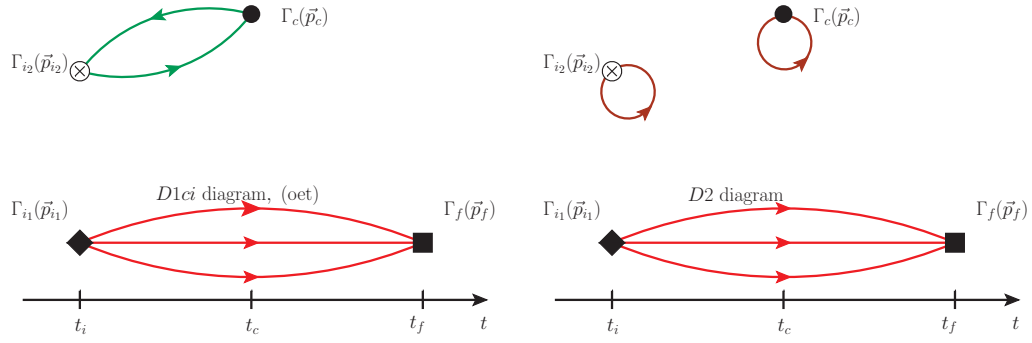


Figure 4: $D1ci$ and $D2$ diagram types with nomenclature used in the contraction formulas.

4.2 Two-hadron interpolating field

Clebsch-Gordan addition for $1 \oplus 1/2 \rightarrow (3/2, +1/2)$ is

$$|3/2, +1/2\rangle = \sqrt{\frac{1}{3}} |1, +1\rangle |1/2, -1/2\rangle + \sqrt{\frac{2}{3}} |1, 0\rangle |1/2, +1/2\rangle \quad (83)$$

$$|\pi N, I = 3/2, I^3 = +1/2\rangle = \sqrt{\frac{1}{3}} \pi^+ n + \sqrt{\frac{2}{3}} \pi^0 p$$

with isospin triplet $\pi = (\pi^+, \pi^0, \pi^-)$ and isospin doublet $N = (p, n)$.

Clebsch-Gordan addition for completeness, $1 \oplus 1/2 \rightarrow (1/2, +1/2)$

$$|1/2, +1/2\rangle = \sqrt{\frac{2}{3}} |1, +1\rangle |1/2, -1/2\rangle - \sqrt{\frac{1}{3}} |1, 0\rangle |1/2, +1/2\rangle \quad (84)$$

$$|\pi N, I = 1/2, I^3 = +1/2\rangle = \sqrt{\frac{2}{3}} \pi^+ n - \sqrt{\frac{1}{3}} \pi^0 p.$$

4.3 Proton / neutron interpolators

Concerning isospin, the proton and neutron interpolators built by coupling 3 isospin-1/2 fields to isospin $|1/2, \pm 1/2\rangle$ states. For completeness we list the couplings below. We use interpolators like

$$N_\alpha = \epsilon_{abc} \left[q_1^{ta} \Gamma q_2^b \right] q_{3\alpha}^c = \{\epsilon_{abc} \Gamma_{\gamma\delta}\} q_{1\gamma}^a q_{2\delta}^b q_{3\alpha}^c \quad (85)$$

and for the combination in eq. (85) we write $q_1 q_2 q_3$ for short.

$$\frac{1}{2} \oplus \frac{1}{2} \rightarrow 0$$

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (ud - du) \quad (86)$$

$$\frac{1}{2} \oplus \frac{1}{2} \rightarrow 1$$

$$\begin{aligned} |1, +1\rangle &= uu \\ |1, 0\rangle &= \frac{1}{\sqrt{2}} (ud + du) \\ |1, -1\rangle &= dd \end{aligned} \quad (87)$$

$$0 \oplus \frac{1}{2} \rightarrow \frac{1}{2}$$

$$\begin{aligned} |1/2, +1/2\rangle &= \frac{1}{\sqrt{2}} (ud - du) u \\ |1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} (ud - du) d \end{aligned} \quad (88)$$

$$1 \oplus \frac{1}{2} \rightarrow \frac{1}{2}$$

$$\begin{aligned} |1/2, +1/2\rangle &= \sqrt{\frac{2}{3}} uud - \sqrt{\frac{1}{6}} (ud + du) u \\ |1/2, -1/2\rangle &= -\sqrt{\frac{2}{3}} ddu + \sqrt{\frac{1}{6}} (ud + du) d \end{aligned} \quad (89)$$

$$1 \oplus \frac{1}{2} \rightarrow \frac{3}{2}$$

$$\begin{aligned} |3/2, +3/2\rangle &= uuu \\ |3/2, +1/2\rangle &= \sqrt{\frac{1}{3}} uud + \sqrt{\frac{1}{3}} (ud + du) u \\ |3/2, -1/2\rangle &= \sqrt{\frac{1}{3}} ddu + \sqrt{\frac{1}{3}} (ud + du) d \\ |3/2, -3/2\rangle &= ddd \end{aligned} \quad (90)$$

Combination with interpolator We use the same sequence quark fields to assign flavors in eq. (85).

From that we get 2 sets of isospin-1/2 interpolators (up to overall normalization).

$$\begin{aligned} N_1^{(\frac{1}{2}, +\frac{1}{2})} &= \epsilon_{abc} \left\{ \left[u^{at} \Gamma d^b \right] u^c - \left[d^{at} \Gamma u^b \right] u^c \right\} \\ N_1^{(\frac{1}{2}, -\frac{1}{2})} &= \epsilon_{abc} \left\{ \left[u^{at} \Gamma d^b \right] u^c - \left[d^{at} \Gamma u^b \right] d^c \right\} \end{aligned} \quad (91)$$

$$\begin{aligned} N_2^{(\frac{1}{2}, +\frac{1}{2})} &= \epsilon_{abc} \left\{ \sqrt{2} \left[u^{at} \Gamma u^b \right] d^c - \frac{1}{\sqrt{2}} \left[u^{at} \Gamma d^b \right] u^c - \frac{1}{\sqrt{2}} \left[d^{at} \Gamma u^b \right] u^c \right\} \\ N_2^{(\frac{1}{2}, -\frac{1}{2})} &= \epsilon_{abc} \left\{ -\sqrt{2} \left[d^{at} \Gamma d^b \right] u^c + \frac{1}{\sqrt{2}} \left[u^{at} \Gamma d^b \right] d^c + \frac{1}{\sqrt{2}} \left[d^{at} \Gamma u^b \right] d^c \right\} \end{aligned} \quad (92)$$

Moreover, we have the isospin-3/2 interpolator

$$\begin{aligned} B^{(\frac{3}{2}, +\frac{3}{2})} &= \epsilon_{abc} \left[u^{at} \Gamma u^b \right] u^c \\ B^{(\frac{3}{2}, +\frac{1}{2})} &= \epsilon_{abc} \left\{ \left[u^{at} \Gamma u^b \right] d^c + \left[u^{at} \Gamma d^b \right] u^c + \left[d^{at} \Gamma u^b \right] u^c \right\} \\ B^{(\frac{3}{2}, -\frac{1}{2})} &= \epsilon_{abc} \left\{ \left[d^{at} \Gamma d^b \right] u^c + \left[u^{at} \Gamma d^b \right] d^c + \left[d^{at} \Gamma u^b \right] d^c \right\} \\ B^{(\frac{3}{2}, -\frac{3}{2})} &= \epsilon_{abc} \left[d^{at} \Gamma d^b \right] d^c \end{aligned} \quad (93)$$

Reduction of basis states for isospin 1/2 We have two orthogonal basis states for $|1/2, +1/2\rangle$. We want to couple $|1/2, +1/2\rangle$ and $|1, +1\rangle$ from the pion triplet to $|3/2, +3/2\rangle$.

We can simplify the interpolators for specific choices of Γ using

$$\begin{aligned} N(q_1 q_2 q_3) &= \epsilon_{abc} \left[q_1^{ta} \Gamma_{\gamma\delta} q_2^b \right] u^c \\ &= -\epsilon_{abc} \left[q_2^{tb} \Gamma_{\gamma\delta} q_1^a \right] u^c \\ &= +\epsilon_{bac} \left[q_2^{tb} \Gamma_{\delta\gamma}^t q_1^a \right] u^c \\ &= \sigma_{\Gamma^t} \epsilon_{abc} \left[q_2^{ta} \Gamma q_1^b \right] u^c \\ &= \sigma_{\Gamma^t} N(q_2 q_1 q_3) \end{aligned} \quad (94)$$

where the first minus sign in the second line is due to q s being Grassmann fields and σ_{Γ^t} is the sign of Γ under transposition (in spinor indices)

$$\Gamma^t = \sigma_{\Gamma^t} \Gamma.$$

We have for $q^t \Gamma q$ scalar

Γ	C	$C\gamma_5$	$C\gamma_5\gamma_0$	$C\gamma_0$
σ_{Γ^t}	-1	-1	-1	$+1$

and for $q^t \Gamma q$ vector $(C\gamma_\mu)^t = C\gamma_\mu$, i.e. $\sigma_{(C\gamma_\mu)^t} = +1$ for $\mu = 0, 1, 2, 3$.

So the proper isospin-1/2 interpolators, depending on Γ are

$$\begin{aligned}
&\text{for } \sigma_{\Gamma^t} = -1 : N_1^{(\frac{1}{2}, +\frac{1}{2})}(\Gamma) = \epsilon_{abc} \left[u^{at} \Gamma d^b \right] u^c \\
&\text{for } \sigma_{\Gamma^t} = +1 : N_2^{(\frac{1}{2}, +\frac{1}{2})}(\Gamma) = \epsilon_{abc} \left[u^{at} \Gamma u^b \right] d^c - \left[u^{at} \Gamma d^b \right] u^c
\end{aligned} \tag{95}$$

The isospin-3/2 interpolators, depending on Γ are the usual ones

$$\begin{aligned}
&\text{for } \sigma_{\Gamma^t} = +1 : B^{(\frac{3}{2}, +\frac{3}{2})}(\Gamma) = \epsilon_{abc} \left[u^{at} \Gamma u^b \right] u^c \\
&\text{for } \sigma_{\Gamma^t} = +1 : B^{(\frac{3}{2}, +\frac{1}{2})}(\Gamma) = \epsilon_{abc} \left[u^{at} \Gamma u^b \right] d^c + 2\epsilon_{abc} \left[u^{at} \Gamma d^b \right] u^c. \\
&\text{for } \sigma_{\Gamma^t} = -1 : B^{(\frac{3}{2}, +\frac{3}{2})}(\Gamma) = 0 \\
&\text{for } \sigma_{\Gamma^t} = -1 : B^{(\frac{3}{2}, +\frac{1}{2})}(\Gamma) = 0
\end{aligned} \tag{96}$$

Comments At the moment, we are using only the nucleon interpolator of type $[u^t \Gamma d] u$ for any Γ . We see from eqs. (95), that this vector has overlap with the properly projected isospin-1/2 vector in case of both Γ symmetric and anti-symmetric under transposition.

We see, that our usage of $[u^t \Gamma d] u$ is the properly projected isospin-1/2 interpolator for $\Gamma = C, C\gamma_5, C\gamma_5\gamma_0$.

In the symmetric case ($\Gamma = C\gamma_0$), the interpolator does not represent a pure isospin-1/2 state, but it is a linear combination of $|1/2, +1/2\rangle$ and $|3/2, +1/2\rangle$,

$$[u^t \Gamma d] u = \frac{1}{\sqrt{3}} B^{(\frac{3}{2}, +\frac{1}{2})} - \frac{1}{\sqrt{6}} N_2^{(\frac{1}{2}, +\frac{1}{2})}.$$

For the 2-point functions we couple this interpolator with the $|1, +1\rangle$ interpolator from the pion-triplet. Thus the third isospin component will be $I^3 = +3/2$. The resulting state will have overlap with $|3/2, +3/2\rangle$ and $|I, +3/2\rangle$ for $I \geq 5/2$.

To get isospin 5/2 we need to couple at least one nucleon and 2 pions ($1/2 \oplus 1 \oplus 1$), which means a state with isospin 5/2 would necessarily mean an $N\pi\pi$ state. For that realm of the spectrum we cannot do elastic 2-particle scattering Lüscher analysis anyway.

4.4 Further comments

Wigner-Eckart like calculation. If we do not want to include both components in the sum on the right-hand side of eq. (83) we use

$$\langle f, I_f, I_f^3 | J, I_c, I_c^3 | i, I_i, I_i^3 \rangle = \langle f, I_f || J, I_c || i, I_i \rangle C_{CG} (I_f, I_f^3 | I_c, I_c^3; I_i, I_i^3) \quad (97)$$

with a Clebsch-Gordan coefficient and reduced matrix element $\langle f || J || i \rangle$, which contains the $N - J - \pi N / \Delta$ coupling stripped of any dependence on specific 3rd isospin component combinations.

Isoscalar and isovector current, initial pion We insert the electromagnetic current J^{em} , whose up-down contribution is linear combination of isoscalar and isovector current. $J^{\text{em}} = \frac{1}{6} J^{I=0, I^3=0} + \frac{1}{2} J^{I=1, I^3=0}$. The final state should be isospin-1/2 (quantum numbers of the nucleon), while the initial state has isospin-3/2 (quantum numbers of the Delta).

Assuming that $SU(2)$ isospin is a symmetry, we must have:

- there is *only a contribution from the light isovector current*; the virtual photon must be isovector;
- hence there are *no disconnected diagrams from the current insertion*, i.e. no contribution from diagrams of type $D1cc$ or any $D2$

In addition, if a neutral pion field is part of the initial time interpolator

- there are *no loop contributions from the neutral pion*, i.e. no contribution from the diagrams of type $D1ii$ (or any $D2$)

Thus *the only (singly) quark-disconnected contributions* arise from the diagrams of type $D1ci$.

4.5 Summary on Nucleon and Delta interpolators

Interpolators are always taken according to (cf. eq. (85))

$$F(\Gamma_1, \Gamma_2) = \epsilon_{abc} \left[q_1^{ta} \Gamma_1 q_2^b \right] \Gamma_2 q_3^c$$

4.5.1 Nucleon interpolators

Antisymmetric in $[q_1 q_2]$ @ isospin 1/2.

$J = 0$	P	T
$\mathbb{1}$	+	+
γ_0	+	-
γ_5	-	-
$\gamma_0 \gamma_5$	-	+

By the sign for P and T we mean $\gamma_0 \Gamma \gamma_0 = \sigma_{\Gamma^0} \Gamma$ for parity and $\gamma_5 \gamma_0 \Gamma \gamma_0 \gamma_5 = \sigma_{\Gamma^{05}} \Gamma$ for timereversal.

The complete interpolator - including Γ_1 and Γ_2 should always transform $F(t, \vec{x}) \rightarrow \gamma_0 F(t, -\vec{x})$ under parity like the nucleon. Thus acceptable combinations are

	Γ_1	Γ_2	$\sigma_{\Gamma_1^t}$
1	$C\gamma_5$	$\mathbb{1}$	-1
2	$C\gamma_5\gamma_0$	$\mathbb{1}$	-1
3	C	γ_5	-1
4	$C\gamma_0$	γ_5	+1

We use 1, 2 and 3 due to $\sigma_{\Gamma_1^t} = -1$.

4.5.2 Delta interpolators

Symmetric in $[q_1, q_2]$ @ isospin 3/2.

$J = 1$	P	T
γ_i	-	+
$\gamma_i \gamma_0$	-	-
$\gamma_i \gamma_5$	+	-
$\gamma_i \gamma_0 \gamma_5$	+	+

Thus acceptable combinations are

	Γ_1	Γ_2	$\sigma_{\Gamma_1^t}$
1	$C\gamma_i$	$\mathbb{1}$	+1
2	$C\gamma_i\gamma_0$	$\mathbb{1}$	+1
3	$C\gamma_i\gamma_0\gamma_5$	γ_5	+1
4	$C\gamma_i\gamma_5$	γ_5	-1

Thus we can use 1, 2, and 3, since we need Γ_1 symmetric.

Note the different behavior under time reversal.

Beware of the (continuum) spin projectors.

4.6 Contraction for $udu - \bar{u}u - \bar{u}u \times \bar{d}\bar{u}\bar{u}$

4.6.1 B -type diagrams

$$B_3 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (98)$$

$$\begin{aligned} & \mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{uu})_{\sigma_1}^l \\ & \mathcal{V}_2 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\sigma_2 \alpha \beta}^l \end{aligned} \quad (99)$$

$$\begin{aligned} & B_4 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \\ & \mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{uu})_{\sigma_1}^l \\ & \mathcal{V}_4 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\alpha \sigma_2 \beta}^l \end{aligned} \quad (100)$$

$$\begin{aligned} & B_5 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \\ & \mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{uu})_{\beta}^l \\ & \mathcal{V}_2 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\sigma_2 \alpha \sigma_1}^l \end{aligned} \quad (101)$$

$$\begin{aligned} & B_6 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \\ & \mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{uu})_{\beta}^l \\ & \mathcal{V}_4 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\alpha \sigma_2 \sigma_1}^l \end{aligned} \quad (102)$$

4.6.2 W-type diagrams

$$W_5 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (103)$$

$$\begin{aligned} & \mathcal{V}_3 \left(\bar{\phi}^{(t_f)}, \gamma_5 \Gamma_c U \right)_\beta^l \\ & \mathcal{V}_4 \left(\gamma_5 \xi^{(t_f)}, \Gamma_f D, T_{fii}^{uu} \right)_{\alpha \sigma_2 \sigma_1}^l \end{aligned}$$

$$W_6 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (104)$$

$$\begin{aligned} & \mathcal{V}_3 \left(\bar{\phi}^{(t_f)}, \gamma_5 \Gamma_c U \right)_\beta^l \\ & \mathcal{V}_2 \left(\gamma_5 \xi^{(t_f)}, \Gamma_f D, T_{fii}^{uu} \right)_{\sigma_2 \alpha \sigma_1}^l \end{aligned}$$

$$W_7 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (105)$$

$$\begin{aligned} & \mathcal{V}_3 \left(\bar{\phi}^{(t_f)}, \gamma_5 \Gamma_c U \right)_{\sigma_1}^l \\ & \mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_{f_1} D, T_{fii}^{uu} \right)_{\alpha \sigma_2 \beta}^l \end{aligned}$$

$$W_8 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (106)$$

$$\begin{aligned} & \mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c U \right)_{\sigma_1}^l \\ & \mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} D, T_{fii}^{uu} \right)_{\sigma_2 \alpha \beta}^l \end{aligned}$$

4.6.3 Z-type diagrams

$$Z_5 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (107)$$

$$\begin{aligned} & \mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c U \right)_{\sigma_1}^l \\ & \mathcal{V}_2 \left(\phi, \Gamma_{f_1} D, U \right)_{\sigma_2 \alpha \beta}^l \end{aligned}$$

$$Z_6 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (108)$$

$$\begin{aligned} & \mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c U \right)_{\sigma_1}^l \\ & \mathcal{V}_4 \left(\phi, \Gamma_{f_1} D, U \right)_{\alpha \sigma_2 \beta}^l \end{aligned}$$

$$Z_7 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (109)$$

$$\begin{aligned} & \mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c U \right)_\beta^l \\ & \mathcal{V}_2 \left(\phi, \Gamma_{f_1} D, U \right)_{\sigma_2 \alpha \sigma_1}^l \end{aligned}$$

$$Z_8 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (110)$$

$$\begin{aligned} & \mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c U \right)_\beta^l \\ & \mathcal{V}_4 \left(\phi, \Gamma_{f_1} D, U \right)_{\alpha \sigma_2 \sigma_1}^l \end{aligned}$$

4.6.4 $D1cc$ -type diagrams

$$D1cc_1 = \mathcal{T}_2 (U, T_{fii}^{uu}, \Gamma_{f_1} D\Gamma_{i_1}) \times L_{cc}^u (\Gamma_c) \quad (111)$$

$$D1cc_2 = \mathcal{T}_1 (T_{fii}^{uu}, \Gamma_{f_1} D\Gamma_{i_1}, U) \times L_{cc}^u (\Gamma_c) \quad (112)$$

$$D1cc_3 = \mathcal{T}_1 (U, \Gamma_{f_1} D\Gamma_{i_1}, T_{fii}^{uu}) \times L_{cc}^u (\Gamma_c) \quad (113)$$

$$D1cc_4 = \mathcal{T}_2 (T_{fii}^{uu}, \Gamma_{f_1} D\Gamma_{i_1}, U) \times L_{cc}^u (\Gamma_c) \quad (114)$$

4.6.5 $D1ci$ -type diagrams

$$D1ci_1 = L_{ci}^{uu} \times N_1 \quad (115)$$

$$D1ci_2 = L_{ci}^{uu} \times N_2 \quad (116)$$

4.6.6 $D1ii$ -type diagrams

$$D1ii_1 = L_{ii}^{uu} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (117)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_1}^l$$

$$\mathcal{V}_2 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\sigma_2 \alpha \beta}^l$$

$$D1ii_2 = L_{ii}^{uu} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (118)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_1}^l$$

$$\mathcal{V}_4 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\alpha \sigma_2 \beta}^l$$

$$D1ii_3 = L_{ii}^{uu} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (119)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\beta}^l$$

$$\mathcal{V}_2 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\sigma_2 \alpha \sigma_1}^l$$

$$D1ii_4 = L_{ii}^{uu} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (120)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\beta}^l$$

$$\mathcal{V}_4 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\alpha \sigma_2 \sigma_1}^l$$

4.6.7 $D2$ -type diagrams

$$D2_1 = L_{ii}^{uu} \times L_{cc}^{uu} \times N_1 \quad (121)$$

$$D2_2 = L_{ii}^{uu} \times L_{cc}^{uu} \times N_2 \quad (122)$$

4.7 Contraction for $udu - \bar{u}u - \bar{d}d \times \bar{d}\bar{u}\bar{u}$

4.7.1 B -type diagrams

NONE

4.7.2 W -type diagrams

$$W_9 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (123)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_1}^l$$

$$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} T_{fii}^{dd}, U \right)_{\sigma_2 \alpha \beta}^l$$

$$W_{10} = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (124)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_1}^l$$

$$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_{f_1} T_{fii}^{dd}, U \right)_{\alpha \sigma_2 \beta}^l$$

$$W_{11} = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (125)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\beta}^l$$

$$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} T_{fii}^{dd}, U \right)_{\sigma_2 \alpha \sigma_1}^l$$

$$W_{12} = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (126)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\beta}^l$$

$$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_{f_1} T_{fii}^{dd}, U \right)_{\alpha \sigma_2 \sigma_1}^l$$

$$(127)$$

4.7.3 Z -type diagrams

NONE

4.7.4 $D1cc$ -type diagrams

$$D1cc_5 = \mathcal{T}_1 \left(U, \Gamma_{f_1} T_{fii}^{dd} \Gamma_{i_1}, U \right) \times L_{cc}^{uu} \quad (128)$$

$$D1cc_6 = \mathcal{T}_2 \left(U, \Gamma_{f_1} T_{fii}^{dd} \Gamma_{i_1}, U \right) \times L_{cc}^{uu} \quad (129)$$

$$(130)$$

4.7.5 $D1ii$ -type diagrams

$$D1ii_5 = L_{ii}^{dd} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (131)$$

$$\begin{aligned} & \mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_1}^l \\ & \mathcal{V}_2 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\sigma_2 \alpha \beta}^l \\ D1ii_6 = & L_{ii}^{dd} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (132) \end{aligned}$$

$$\begin{aligned} & \mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_1}^l \\ & \mathcal{V}_4 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\alpha \sigma_2 \beta}^l \\ D1ii_7 = & L_{ii}^{dd} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (133) \end{aligned}$$

$$\begin{aligned} & \mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\beta}^l \\ & \mathcal{V}_2 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\sigma_2 \alpha \sigma_1}^l \\ D1ii_8 = & L_{ii}^{dd} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (134) \\ & \mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\beta}^l \\ & \mathcal{V}_4 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\alpha \sigma_2 \sigma_1}^l \end{aligned}$$

4.7.6 $D2$ -type diagrams

$$D2_3 = L_{ii}^{dd} \times L_{cc}^{uu} \times N_1 \quad (135)$$

$$D2_4 = L_{ii}^{dd} \times L_{cc}^{uu} \times N_2 \quad (136)$$

4.8 Contraction for $udu - \bar{d}d - \bar{u}u \times \bar{d}\bar{u}\bar{u}$

4.8.1 B -type diagrams

NONE

4.8.2 W -type diagrams

$$W_{13} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \quad (137)$$

$$\mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l$$

$$\mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, T_{fii}^{uu}, U)_{\sigma_1\alpha\beta}^l$$

$$W_{14} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \quad (138)$$

$$\mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l$$

$$\mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, U, T_{fii}^{uu})_{\beta\alpha\sigma_1}^l$$

$$W_{15} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \quad (139)$$

$$\mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l$$

$$\mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, T_{fii}^{uu}, U)_{\beta\alpha\sigma_1}^l$$

$$W_{16} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \quad (140)$$

$$\mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l$$

$$\mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, U, T_{fii}^{uu})_{\sigma_1\alpha\beta}^l$$

$$(141)$$

4.8.3 Z -type diagrams

NONE

4.8.4 $D1cc$ -type diagrams

$$D1cc_7 = \mathcal{T}_1(T_{fii}^{uu}, \Gamma_{f_1} D \Gamma_{i_1}, U) \times L_{cc}^{dd} \quad (142)$$

$$D1cc_8 = \mathcal{T}_2(U, \Gamma_{f_1} D \Gamma_{i_1}, T_{fii}^{uu}) \times L_{cc}^{dd} \quad (143)$$

$$D1cc_9 = \mathcal{T}_1(U, \Gamma_{f_1} D \Gamma_{i_1}, T_{fii}^{uu}) \times L_{cc}^{dd} \quad (144)$$

$$D1cc_{10} = \mathcal{T}_2(T_{fii}^{uu}, \Gamma_{f_1} D \Gamma_{i_1}, U) \times L_{cc}^{dd} \quad (145)$$

4.8.5 $D1ii$ -type diagrams

$$D1ii_9 = L_{ii}^{uu} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (146)$$

$$\begin{aligned} & \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l \\ & \mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, U, U)_{\beta \alpha \sigma_1}^l \\ D1ii_{10} &= L_{ii}^{uu} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (147) \\ & \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l \\ & \mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, U, U)_{\sigma_1 \alpha \beta}^l \end{aligned}$$

$$(148)$$

4.8.6 $D2$ -type diagrams

$$D2_5 = L_{ii}^{uu} \times L_{cc}^{dd} \times N_1 \quad (149)$$

$$D2_6 = L_{ii}^{uu} \times L_{cc}^{dd} \times N_2 \quad (150)$$

4.9 Contraction for $udu - \bar{d}d - \bar{d}d \times \bar{d}\bar{u}\bar{u}$

4.9.1 B -type diagrams

$$B_7 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (151)$$

$$\begin{aligned} & \mathcal{V}_3\left(\phi, \gamma_5 \Gamma_c T_{fii}^{dd}\right)_{\sigma_2}^l \\ & \mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, U, U)_{\beta \alpha \sigma_1}^l \\ B_8 &= (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (152) \\ & \mathcal{V}_3\left(\phi, \gamma_5 \Gamma_c T_{fii}^{dd}\right)_{\sigma_2}^l \\ & \mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, U, U)_{\sigma_1 \alpha \beta}^l \end{aligned}$$

4.9.2 W -type diagrams

NONE

4.9.3 Z-type diagrams

$$Z_9 = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (153)$$

$$\begin{aligned} & \mathcal{V}_3 (\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l \\ & \mathcal{V}_2 (\Gamma_{f_1} \bar{\phi}, U, U)_{\beta \alpha \sigma_1}^l \\ Z_{10} &= (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \end{aligned} \quad (154)$$

$$\begin{aligned} & \mathcal{V}_3 (\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l \\ & \mathcal{V}_2 (\Gamma_{f_1} \bar{\phi}, U, U)_{\sigma_1 \alpha \beta}^l \end{aligned} \quad (155)$$

4.9.4 D1cc-type diagrams

$$D1cc_{11} = \mathcal{T}_1 \left(U, \Gamma_{f_1} T_{fii}^{dd} \Gamma_{i_1}, U \right) \times L_{cc}^{dd} \quad (156)$$

$$D1cc_{12} = \mathcal{T}_2 \left(U, \Gamma_{f_1} T_{fii}^{dd} \Gamma_{i_1}, U \right) \times L_{cc}^{dd} \quad (157)$$

4.9.5 D1ci-type diagrams

$$D1ci_3 = L_{ci}^{dd} \times N_1 \quad (158)$$

$$D1ci_4 = L_{ci}^{dd} \times N_2 \quad (159)$$

4.9.6 D1ii-type diagrams

$$D1ii_{11} = L_{ii}^{dd} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (160)$$

$$\begin{aligned} & \mathcal{V}_3 (\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l \\ & \mathcal{V}_2 (\Gamma_{f_1} \gamma_5 \xi, U, U)_{\beta \alpha \sigma_1}^l \\ D1ii_{12} &= L_{ii}^{dd} \times (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \end{aligned} \quad (161)$$

$$\begin{aligned} & \mathcal{V}_3 (\phi, \gamma_5 \Gamma_c D)_{\sigma_2}^l \\ & \mathcal{V}_2 (\Gamma_{f_1} \gamma_5 \xi, U, U)_{\sigma_1 \alpha \beta}^l \end{aligned} \quad (162)$$

4.9.7 D2-type diagrams

$$D2_7 = L_{ii}^{dd} \times L_{cc}^{dd} \times N_1 \quad (163)$$

$$D2_8 = L_{ii}^{dd} \times L_{cc}^{dd} \times N_2 \quad (164)$$

4.10 Contraction for $udu - \bar{u}u - \bar{u}d \times \bar{u}d\bar{d}$

4.10.1 B-type diagrams

$$B_9 = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \quad (165)$$

$$\mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{ud} \right)_{\sigma_1}^l$$

$$\mathcal{V}_2 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\beta\alpha\sigma_2}^l$$

$$B_{10} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \quad (166)$$

$$\mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{ud} \right)_{\sigma_1}^l$$

$$\mathcal{V}_4 (\gamma_5 \xi, U, \Gamma_{f_1} D)_{\alpha\sigma_2\beta}^l$$

$$B_{11} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \quad (167)$$

$$\mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{ud} \right)_{\beta}^l$$

$$\mathcal{V}_2 (\gamma_5 \xi, \Gamma_{f_1} D, U)_{\sigma_1\alpha\sigma_2}^l$$

$$B_{12} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \quad (168)$$

$$\mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{ud} \right)_{\beta}^l$$

$$\mathcal{V}_4 (\gamma_5 \xi, U, \Gamma_{f_1} D)_{\alpha\sigma_2\sigma_1}^l$$

4.10.2 *W*-type diagrams

$$W_{17} = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (169)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_2}^l$$

$$\mathcal{V}_4 \left(\gamma_5 \xi, T_{fii}^{ud}, \Gamma_{f_1} D \right)_{\alpha \sigma_1 \beta}^l$$

$$W_{18} = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (170)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_2}^l$$

$$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} D, T_{fii}^{ud} \right)_{\beta \alpha \sigma_1}^l$$

$$W_{19} = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (171)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_2}^l$$

$$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_{f_1} D, T_{fii}^{ud} \right)_{\alpha \sigma_1 \beta}^l$$

$$W_{20} = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (172)$$

$$\mathcal{V}_3 (\bar{\phi}, \gamma_5 \Gamma_c U)_{\sigma_2}^l$$

$$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} D, T_{fii}^{ud} \right)_{\sigma_1 \alpha \beta}^l$$

4.10.3 *Z*-type diagrams

NONE

4.10.4 *D1cc*-type diagrams

$$D1cc_{13} = L_{cc}^{uu} \times \mathcal{T}_1 \left(U \Gamma_{i_1}, T_{fii}^{ud}, \Gamma_{f_1} D \right) \quad (173)$$

$$D1cc_{14} = L_{cc}^{uu} \times \mathcal{T}_1 \left(T_{fii}^{ud}, U \Gamma_{i_1}, \Gamma_{f_1} D \right) \quad (174)$$

$$D1cc_{15} = L_{cc}^{uu} \times \mathcal{T}_1 \left(U \Gamma_{i_1}, \Gamma_{f_1} D, T_{fii}^{ud} \right) \quad (175)$$

$$D1cc_{16} = L_{cc}^{uu} \times \mathcal{T}_2 \left(T_{fii}^{ud}, \Gamma_{f_1} D, U \Gamma_{i_1} \right) \quad (176)$$

4.10.5 *D1ii*-type diagrams

NONE

4.10.6 *D1ci*-type diagrams

NONE

4.10.7 D2-type diagrams

NONE

4.11 Contraction for $udu - \bar{d}d - \bar{u}d \times \bar{u}d\bar{d}$

4.11.1 B-type diagrams

NONE

4.11.2 W-type diagrams

$$W_{21} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_\beta^l \mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, T_{fii}^{ud}, U)_{\sigma_1\alpha\sigma_2} \quad (177)$$

$$W_{22} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_\beta^l \mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, U, T_{fii}^{ud})_{\sigma_2\alpha\sigma_1}^l \quad (178)$$

$$W_{23} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_1}^l \mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, T_{fii}^{ud}, U)_{\beta\alpha\sigma_2}^l \quad (179)$$

$$W_{24} = (-1) (\Gamma_{i_1})_{\sigma_1\sigma_2}^t \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_1}^l \mathcal{V}_2(\Gamma_{f_1} \gamma_5 \xi, U, T_{fii}^{ud})_{\sigma_2\alpha\beta}^l \quad (180)$$

4.11.3 Z-type diagrams

$$Z_{11} = (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \quad (181)$$

$$\begin{aligned} & \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_1}^l \\ & \mathcal{V}_2(\phi, \Gamma_{f_1} D, U)_{\beta \alpha \sigma_2}^l \\ Z_{12} &= (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \end{aligned} \quad (182)$$

$$\begin{aligned} & \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\sigma_1}^l \\ & \mathcal{V}_4(\phi, U, \Gamma_{f_1} D)_{\alpha \sigma_2 \beta}^l \\ Z_{13} &= (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \end{aligned} \quad (183)$$

$$\begin{aligned} & \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\beta}^l \\ & \mathcal{V}_2(\phi, \Gamma_{f_1} D, U)_{\sigma_1 \alpha \sigma_2}^l \\ Z_{14} &= (-1) (\Gamma_{i_1})_{\sigma_1 \sigma_2}^t \end{aligned} \quad (184)$$

$$\begin{aligned} & \mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)_{\beta}^l \\ & \mathcal{V}_4(\phi, U, \Gamma_{f_1} D)_{\alpha \sigma_2 \sigma_1}^l \end{aligned}$$

4.11.4 D1cc-type diagrams

$$D1cc_{17} = L_{cc}^{dd} \times \mathcal{T}_1 \left(U \Gamma_{i_1}, T_{fii}^{ud}, \Gamma_{f_1} D \right) \quad (185)$$

$$D1cc_{18} = L_{cc}^{dd} \times \mathcal{T}_1 \left(T_{fii}^{ud}, U \Gamma_{i_1}, \Gamma_{f_1} D \right) \quad (186)$$

$$D1cc_{19} = L_{cc}^{dd} \times \mathcal{T}_1 \left(U \Gamma_{i_1}, \Gamma_{f_1} D, T_{fii}^{ud} \right) \quad (187)$$

$$D1cc_{20} = L_{cc}^{dd} \times \mathcal{T}_2 \left(T_{fii}^{ud}, \Gamma_{f_1} D, U \Gamma_{i_1} \right) \quad (188)$$

4.11.5 D1ii-type diagrams

NONE

4.11.6 D1ci-type diagrams

NONE

4.12 Delta - current - Nucleon triangle diagrams

4.12.1 $udu - \bar{u}u - \bar{d}\bar{u}\bar{u}$

$$\begin{aligned} \langle udu \bar{u}u \bar{d}\bar{u}\bar{u} \rangle = & -U \Gamma_c U (\Gamma_f D \Gamma_i)^T U - U \Gamma_c U \left[U (\Gamma_f D \Gamma_i)^T \right] \\ & - U (\Gamma_f D \Gamma_i)^T U \Gamma_c U - U \left[U \Gamma_c U (\Gamma_f D \Gamma_i)^T \right] \end{aligned} \quad (189)$$

$$[-U \Gamma_c] \left(-U (\Gamma_f D \Gamma_i)^T U - U \left[U (\Gamma_f D \Gamma_i)^T \right] \right) \quad (190)$$

$$\begin{aligned} = & -(\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_4(\phi_u, \Gamma_{f1} D, U)_{\alpha, \sigma_2, \beta}^l \mathcal{V}_3(\xi, \Gamma_c U)_{\sigma_1}^l \\ & -(\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_4(\phi_u, \Gamma_{f1} D, U)_{\alpha, \sigma_2, \sigma_1}^l \mathcal{V}_3(\xi, \Gamma_c U)_{\beta}^l \\ & -(\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_2(\phi_u, \Gamma_{f1} D, U)_{\sigma_2, \alpha, \sigma_1}^l \mathcal{V}_3(\xi, \Gamma_c U)_{\beta}^l \\ & -(\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_2(\phi_u, \Gamma_{f1} D, U)_{\sigma_2, \alpha, \beta}^l \mathcal{V}_3(\xi, \Gamma_c U)_{\sigma_1}^l \end{aligned}$$

$$+ L_c^u \times (\mathcal{T}_1(U, \Gamma_{f1} D \Gamma_{i1}, U) + \mathcal{T}_2(U, \Gamma_{f1} D \Gamma_{i1}, U))$$

4.12.2 $udu - \bar{d}\bar{d} - \bar{d}\bar{u}\bar{u}$

$$\langle udu \bar{d}\bar{d} \bar{d}\bar{u}\bar{u} \rangle = -U (\Gamma_f D \Gamma_c D \Gamma_i)^T U - U \left[U (\Gamma_f D \Gamma_c D \Gamma_i)^T \right] \quad (191)$$

$$[-D \Gamma_c] \left(-U (\Gamma_f D \Gamma_i)^T U - U \left[U (\Gamma_f D \Gamma_i)^T \right] \right) \quad (192)$$

$$\begin{aligned} = & -(\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_2(\phi_d, \Gamma_{f1}^T U, U)_{\beta, \alpha, \sigma_1}^l \mathcal{V}_3(\xi, \Gamma_c D)_{\sigma_2}^l \\ & -(\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_2(\phi_d, \Gamma_{f1}^T U, U)_{\sigma_1, \alpha, \beta}^l \mathcal{V}_3(\xi, \Gamma_c D)_{\sigma_2}^l \end{aligned}$$

$$+ L_c^d \times (\mathcal{T}_1(U, \Gamma_{f1} D \Gamma_{i1}, U) + \mathcal{T}_2(U, \Gamma_{f1} D \Gamma_{i1}, U))$$

4.12.3 $udu - \bar{u}u - \bar{u}\bar{u}\bar{d}$

$$\begin{aligned} \langle udu \bar{u}u \bar{u}\bar{d} \rangle &= -U \Gamma_c U \Gamma_i U^T \Gamma_f D - U \Gamma_c U \Gamma_i^T U^T \Gamma_f D \\ &\quad - U \Gamma_i (U \Gamma_c U)^T \Gamma_f D - U \Gamma_i^T (U \Gamma_c U)^T \Gamma_f D \end{aligned} \quad (193)$$

$$+ [U \Gamma_c] (U \Gamma_i U^T \Gamma_f D + U \Gamma_i^T U^T \Gamma_f D) \quad (194)$$

$$\begin{aligned} &= -(\Gamma_{i1})_{\sigma_1 \sigma_2} \mathcal{V}_4(\phi_u, \Gamma_{f1}^T U, D)_{\alpha, \sigma_2, \beta}^l \mathcal{V}_3(\xi, \Gamma_c U)_{\sigma_1}^l \\ &\quad - (\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_4(\phi_u, \Gamma_{f1}^T U, D)_{\alpha, \sigma_2, \beta}^l \mathcal{V}_3(\xi, \Gamma_c U)_{\sigma_1}^l \\ &\quad - (\Gamma_{i1})_{\sigma_1 \sigma_2} \mathcal{V}_2(\phi_u, \Gamma_{f1} D, U)_{\beta, \alpha, \sigma_1}^l \mathcal{V}_3(\xi, \Gamma_c U)_{\sigma_2}^l \\ &\quad - (\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_2(\phi_u, \Gamma_{f1} D, U)_{\beta, \alpha, \sigma_1}^l \mathcal{V}_3(\xi, \Gamma_c U)_{\sigma_2}^l \\ &\quad + L_c^u \times (\mathcal{T}_1(U, \Gamma_{f1}^T U \Gamma_{i1}^T, D) + \mathcal{T}_2(U, \Gamma_{f1}^T U \Gamma_{i1}, D)) \end{aligned}$$

4.12.4 $udu - \bar{d}d - \bar{u}\bar{u}\bar{d}$

$$\langle udu \bar{d}d \bar{u}\bar{u}\bar{d} \rangle = -U \Gamma_i U^T \Gamma_f D \Gamma_c D - U \Gamma_i^T U^T \Gamma_f D \Gamma_c D \quad (195)$$

$$+ [D \Gamma_c] (U \Gamma_i U^T \Gamma_f D + U \Gamma_i^T U^T \Gamma_f D) \quad (196)$$

$$\begin{aligned} &= -(\Gamma_{i1})_{\sigma_1 \sigma_2} \mathcal{V}_2(\phi_d, \Gamma_{f1}^T U, U)_{\sigma_2, \alpha, \sigma_1}^l \mathcal{V}_3(\xi, \Gamma_c D)_{\beta}^l \\ &\quad - (\Gamma_{i1}^T)_{\sigma_1 \sigma_2} \mathcal{V}_2(\phi_d, \Gamma_{f1}^T U, U)_{\sigma_2, \alpha, \sigma_1}^l \mathcal{V}_3(\xi, \Gamma_c D)_{\beta}^l \\ &\quad + L_c^d \times (\mathcal{T}_1(U, \Gamma_{f1}^T U \Gamma_{i1}^T, D) + \mathcal{T}_2(U, \Gamma_{f1}^T U \Gamma_{i1}, D)) \end{aligned}$$

4.13 Conclusion I - minimal set of reduction operations

The minimal set of reduction we need is given by

reduction type	reduction op	diagrams
$\mathcal{V}_3$	$\mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{uu} \right)$	$B_3^{3pt}, B_4^{3pt}, B_5^{3pt}, B_6^{3pt},$
	$\mathcal{V}_3 \left(\phi, \gamma_5 \Gamma_c T_{fii}^{dd} \right)$	$B_7^{3pt}, B_8^{3pt},$
	$\mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c T_{fii}^{ud} \right)$	$B_9^{3pt}, B_{10}^{3pt}, B_{11}^{3pt}, B_{12}^{3pt}$
	$\mathcal{V}_3 \left(\bar{\phi}, \gamma_5 \Gamma_c U \right)$	$W_5^{3pt}, W_6^{3pt}, W_7^{3pt}, W_8^{3pt},$ $D1i_1^{3pt}, D1i_2^{3pt}, D1i_3^{3pt}, D1i_4^{3pt}$ $W_9^{3pt}, W_{10}^{3pt}, W_{11}^{3pt}, W_{12}^{3pt}$ $W_{17}^{3pt}, W_{18}^{3pt}, W_{19}^{3pt}, W_{20}^{3pt}$ $D1i_5^{3pt}, D1i_6^{3pt}, D1i_7^{3pt}, D1i_8^{3pt}$
	$\mathcal{V}_3 \left(\phi, \gamma_5 \Gamma_c D \right)$	$W_{13}^{3pt}, W_{14}^{3pt}, W_{15}^{3pt}, W_{16}^{3pt},$ $W_{21}^{3pt}, W_{22}^{3pt}, W_{23}^{3pt}, W_{24}^{3pt},$ $D1i_9^{3pt}, D1i_{10}^{3pt},$ $D1i_{11}^{3pt}, D1i_{12}^{3pt}$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f D, U \right)$	$B_3^{3pt}, B_5^{3pt}, B_9^{3pt}, B_{11}^{3pt}$ $D1i_1, D1i_3, D1i_5, D1i_7$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f^t U, U \right)$	$D1i_9^{3pt}, D1i_{10}^{3pt},$ $D1i_{11}^{3pt}, D1i_{12}^{3pt}$ $B_7^{3pt}, B_8^{3pt},$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f D, T_{fii}^{uu} \right)$	W_6^{3pt}, W_8^{3pt}
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f D, T_{fii}^{ud} \right)$	$W_{18}^{3pt}, W_{20}^{3pt}$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f^t U, T_{fii}^{uu} \right)$	$W_{14}^{3pt}, W_{16}^{3pt}$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f^t U, T_{fii}^{ud} \right)$	$W_{22}^{3pt}, W_{24}^{3pt}$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f T_{fii}^{dd}, U \right)$	W_9^{3pt}, W_{11}^{3pt}
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f^t T_{fii}^{uu}, U \right)$	$W_{13}^{3pt}, W_{15}^{3pt}$
$\mathcal{V}_2$	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_f^t T_{fii}^{ud}, U \right)$	$W_{21}^{3pt}, W_{23}^{3pt}$
$\mathcal{V}_4$	$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_f D, U \right)$	$B_4^{3pt}, B_6^{3pt},$ $D1i_2^{3pt}, D1i_4^{3pt}, D1i_6^{3pt}, D1i_8^{3pt}$
$\mathcal{V}_4$	$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_f D, T_{fii}^{uu} \right)$	W_5^{3pt}, W_7^{3pt}
$\mathcal{V}_4$	$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_f D, T_{fii}^{ud} \right)$	$W_{17}^{3pt}, W_{19}^{3pt}$
$\mathcal{V}_4$	$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_f T_{fii}^{dd}, U \right)$	$W_{10}^{3pt}, W_{12}^{3pt}$
$\mathcal{V}_4$	$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_f^t U, D \right)$	$B_{10}^{3pt}, B_{12}^{3pt}$

reduction type	reduction op	diagrams
$\mathcal{V}_3$	$\mathcal{V}_3(\bar{\phi}, \gamma_5 \Gamma_c U)$	$Z_5^{3pt}, Z_6^{3pt}, Z_7^{3pt}, Z_8^{3pt}$
$\mathcal{V}_3$	$\mathcal{V}_3(\phi, \gamma_5 \Gamma_c D)$	$Z_9^{3pt}, Z_{10}^{3pt}, Z_{11}^{3pt}, Z_{12}^{3pt}, Z_{13}^{3pt}, Z_{14}^{3pt},$
$\mathcal{V}_2$	$\mathcal{V}_2(\phi, \Gamma_f D, U)$	$Z_5^{3pt}, Z_7^{3pt}, Z_{11}^{3pt}, Z_{13}^{3pt},$
$\mathcal{V}_4$	$\mathcal{V}_4(\phi, \Gamma_f D, U)$	$Z_6^{3pt}, Z_8^{3pt}, Z_{12}^{3pt}, Z_{14}^{3pt}$

4.14 Conclusion II - minimal diagram construction for 3-point functions

flavor	diag	F 2/4	F 3	OP_{i_1}	OP_{f_1}	$OP_{\mathcal{D}}$	π	IdW	Idtm
$udu - \bar{u}u - \bar{d}\bar{u}\bar{u}$	$B_3^{3\text{pt}}$	$\mathcal{V}_2(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\xi, \Gamma_c T_{fi}^{uu})$	N	N	N	1342	$B_3^{3\text{pt}}$	$\tilde{B}_3^{3\text{pt}}$
	$B_5^{3\text{pt}}$			N	N	N	1324	$B_5^{3\text{pt}}$	$\tilde{B}_5^{3\text{pt}}$
	$B_4^{3\text{pt}}$	$\mathcal{V}_4(\phi_u, \Gamma_{f_1} D, U)$		N	N	N	3142	$B_4^{3\text{pt}}$	$\tilde{B}_4^{3\text{pt}}$
	$B_6^{3\text{pt}}$			N	N	N	3124	$B_6^{3\text{pt}}$	$\tilde{B}_6^{3\text{pt}}$
$udu - \bar{d}\bar{d} - \bar{d}\bar{d} - \bar{d}\bar{u}\bar{u}$	$B_7^{3\text{pt}}$	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_c T_{fi}^{dd})$	T	T	T	3412	$B_7^{3\text{pt}}$	$\tilde{B}_7^{3\text{pt}}$
	$B_8^{3\text{pt}}$			T	T	N	1342	$B_3^{3\text{pt}}$	$\tilde{B}_8^{3\text{pt}}$
$udu - \bar{u}u - \bar{u}d - \bar{u}d\bar{d}$	$B_9^{3\text{pt}}$	$\mathcal{V}_2(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\xi, \Gamma_c T_{fi}^{ud})$	N	N	T	3412	$B_7^{3\text{pt}}$	$\tilde{B}_9^{3\text{pt}}$
	$B_{11}^{3\text{pt}}$			T	N	N	1324	$B_5^{3\text{pt}}$	$\tilde{B}_{11}^{3\text{pt}}$
	$B_{10}^{3\text{pt}}$	$\mathcal{V}_4(\phi_u, \Gamma_{f_1}^T U, D)$		N	T	N	3142	$B_4^{3\text{pt}}$	$\tilde{B}_{10}^{3\text{pt}}$
	$B_{12}^{3\text{pt}}$			N	T	N	3124	$B_6^{3\text{pt}}$	$\tilde{B}_{12}^{3\text{pt}}$

flavor	diag	F 2/4	F 3	OP_{t_1}	OP_{f_1}	$OP_{\mathcal{D}}$	π	IdW	Idtm
$udu - \bar{u}u - \bar{d}u\bar{u}$	W_5^{3pt}	$\mathcal{V}_4(\gamma_5\xi, \Gamma_{f_1}D, T_{f\bar{u}}^{uu})$	$\mathcal{V}_3(\phi_d, \gamma_5\Gamma_c U)$	N	N	N	3124	W_5^{3pt}	$\tilde{W}_5^{3pt}$
	W_7^{3pt}			N	N	N	3142	W_7^{3pt}	$\tilde{W}_7^{3pt}$
	W_6^{3pt}	$\mathcal{V}_2(\gamma_5\xi, \Gamma_{f_1}D, T_{f\bar{u}}^{uu})$		N	N	N	1324	W_6^{3pt}	$\tilde{W}_6^{3pt}$
	W_8^{3pt}			N	N	N	1342	W_8^{3pt}	$\tilde{W}_8^{3pt}$
$udu - \bar{u}u - \bar{d}d - \bar{d}u\bar{u}$	W_9^{3pt}	$\mathcal{V}_2(\gamma_5\xi, \Gamma_{f_1}T_{f\bar{u}}^{dd}, U)$		N	N	N	1342	W_9^{3pt}	$\tilde{W}_9^{3pt}$
	W_{11}^{3pt}			N	N	N	1324	W_{11}^{3pt}	$\tilde{W}_{11}^{3pt}$
	W_{10}^{3pt}	$\mathcal{V}_4(\gamma_5\xi, \Gamma_{f_1}^T U, T_{f\bar{u}}^{dd})$		N	T	N	3412	W_{10}^{3pt}	$\tilde{W}_{10}^{3pt}$
	W_{12}^{3pt}			T	T	N	3124	W_5^{3pt}	$\tilde{W}_{12}^{3pt}$
$udu - \bar{d}d - \bar{u}u - \bar{d}u\bar{u}$	W_{13}^{3pt}	$\mathcal{V}_2(\gamma_5\xi, \Gamma_{f_1}^T T_{f\bar{u}}^{uu}, U)$	$\mathcal{V}_3(\phi_u, \gamma_5\Gamma_c D)$	T	T	N	1342	W_9^{3pt}	$\tilde{W}_{13}^{3pt}$
	W_{15}^{3pt}			T	T	T	3412	W_{15}^{3pt}	$\tilde{W}_{15}^{3pt}$
	W_{14}^{3pt}	$\mathcal{V}_2(\gamma_5\xi, \Gamma_{f_1}^T U, T_{f\bar{u}}^{uu})$		T	T	T	3412	W_{14}^{3pt}	$\tilde{W}_{14}^{3pt}$
	W_{16}^{3pt}			T	T	N	1342	W_8^{3pt}	$\tilde{W}_{16}^{3pt}$

flavor	diag	F 2/4	F 3	OP_{i_1}	OP_{f_1}	OP_D	π	IdW	Idtm
$udu - \bar{u}u - \bar{u}d - \bar{u}d\bar{d}$	W_{17}^{3pt}	$\mathcal{V}_4 \left(\gamma_5 \xi, \Gamma_{f_1} D, T_{f\bar{u}}^{ud} \right)$	$\mathcal{V}_3 (\phi_d, \gamma_5 \Gamma_c U)$	T	N	N	3412	W_{10}^{3pt}	$\tilde{W}_{17}^{3pt}$
	W_{19}^{3pt}			T	N	N	3142	W_7^{3pt}	$\tilde{W}_{19}^{3pt}$
	W_{18}^{3pt}	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1} D, T_{f\bar{u}}^{ud} \right)$		T	N	T	3412	W_{14}^{3pt}	$\tilde{W}_{18}^{3pt}$
	W_{20}^{3pt}			T	N	N	1342	W_8^{3pt}	$\tilde{W}_{20}^{3pt}$
$udu - \bar{d}d - \bar{u}d - \bar{u}d\bar{d}$	W_{21}^{3pt}	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1}^T T_{f\bar{u}}^{ud}, U \right)$	$\mathcal{V}_3 (\phi_u, \gamma_5 \Gamma_c D)$	T	T	N	1324	W_{11}^{3pt}	$\tilde{W}_{21}^{3pt}$
	W_{23}^{3pt}			N	T	T	3412	W_{15}^{3pt}	$\tilde{W}_{23}^{3pt}$
	W_{22}^{3pt}	$\mathcal{V}_2 \left(\gamma_5 \xi, \Gamma_{f_1}^T U, T_{f\bar{u}}^{ud} \right)$		N	T	N	1324	W_6^{3pt}	$\tilde{W}_{22}^{3pt}$
	W_{24}^{3pt}			N	T	N	1342	W_8^{3pt}	$\tilde{W}_{24}^{3pt}$

flavor	diag	F 2/4	F 3	OP_{t_1}	OP_{f_1}	$OP_{\mathcal{D}}$	π	IdW	Idtm
$udu - \bar{u}u - \bar{u}u - \bar{d}\bar{u}\bar{u}$	$Z_5^{3\text{pt}}$	$\mathcal{V}_2(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\phi_d, \gamma_5 \Gamma_c U)$	N	N	N	1342	$Z_5^{3\text{pt}}$	$\tilde{Z}_5^{3\text{pt}}$
	$Z_7^{3\text{pt}}$			N	N	N	1324	$Z_7^{3\text{pt}}$	$\tilde{Z}_7^{3\text{pt}}$
	$Z_6^{3\text{pt}}$	$\mathcal{V}_4(\phi_u, \Gamma_{f_1} D, U)$		N	N	N	3142	$Z_6^{3\text{pt}}$	$\tilde{Z}_6^{3\text{pt}}$
	$Z_8^{3\text{pt}}$			N	N	N	3124	$Z_8^{3\text{pt}}$	$\tilde{Z}_8^{3\text{pt}}$
$udu - \bar{d}\bar{d} - \bar{d}\bar{d} - \bar{d}\bar{u}\bar{u}$	$Z_9^{3\text{pt}}$	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_c D)$	T	T	T	3412	$Z_9^{3\text{pt}}$	$\tilde{Z}_9^{3\text{pt}}$
	$Z_{10}^{3\text{pt}}$			T	T	N	1342	$Z_{10}^{3\text{pt}}$	$\tilde{Z}_{10}^{3\text{pt}}$
$udu - \bar{d}\bar{d} - \bar{u}\bar{d} - \bar{u}\bar{d}\bar{d}$	$Z_{11}^{3\text{pt}}$	$\mathcal{V}_2(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\phi_u, \gamma_5 \Gamma_c D)$	N	N	T	3412	$Z_9^{3\text{pt}}$	$\tilde{Z}_{11}^{3\text{pt}}$
	$Z_{13}^{3\text{pt}}$			T	N	N	1324	$Z_7^{3\text{pt}}$	$\tilde{Z}_{13}^{3\text{pt}}$
	$Z_{12}^{3\text{pt}}$	$\mathcal{V}_4(\phi_u, \Gamma_{f_1} D, U)$		N	N	N	3412	$Z_{12}^{3\text{pt}}$	$\tilde{Z}_{12}^{3\text{pt}}$
	$Z_{14}^{3\text{pt}}$			T	N	N	3124	$Z_8^{3\text{pt}}$	$\tilde{Z}_{14}^{3\text{pt}}$

flavor	diag	F 2/4	F 3	OP_{i_1}	OP_{f_1}	$OP_{\mathcal{D}}$	π	IdW	Idtm
$udu - \bar{u}u - \bar{d}\bar{u}\bar{u}$	T_1^{3pt}	$\mathcal{V}_4(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\xi, \Gamma_c U)$	N	N	N	3142	T_1^{3pt}	$\tilde{T}_1^{3pt}$
	T_2^{3pt}			N	N	N	3124	T_2^{3pt}	$\tilde{T}_2^{3pt}$
	T_3^{3pt}			N	N	N	1324	T_3^{3pt}	$\tilde{T}_3^{3pt}$
	T_4^{3pt}	$\mathcal{V}_2(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\xi, \Gamma_c U)$	N	N	N	1342	T_4^{3pt}	$\tilde{T}_4^{3pt}$
$udu - \bar{d}d - \bar{d}\bar{u}\bar{u}$	T_5^{3pt}	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_c D)$	T	T	T	3412	T_5^{3pt}	$\tilde{T}_5^{3pt}$
	T_6^{3pt}			T	T	N	1342	T_4^{3pt}	$\tilde{T}_6^{3pt}$
$udu - \bar{u}u - \bar{u}\bar{u}\bar{d}$	T_7^{3pt}	$\mathcal{V}_4(\phi_u, \Gamma_{f_1}^T U, D)$	$\mathcal{V}_3(\xi, \Gamma_c U)$	T	T	N	3142	T_1^{3pt}	$\tilde{T}_7^{3pt}$
	T_8^{3pt}			N	T	N	3142	T_1^{3pt}	$\tilde{T}_7^{3pt}$
	T_9^{3pt}	$\mathcal{V}_2(\phi_u, \Gamma_{f_1} D, U)$	$\mathcal{V}_3(\xi, \Gamma_c U)$	N	N	T	3412	T_5^{3pt}	$\tilde{T}_9^{3pt}$
	T_{10}^{3pt}			T	N	T	3412	T_5^{3pt}	$\tilde{T}_9^{3pt}$
$udu - \bar{d}d - \bar{u}\bar{u}\bar{d}$	T_{11}^{3pt}	$\mathcal{V}_2(\phi_d, \Gamma_{f_1}^T U, U)$	$\mathcal{V}_3(\xi, \Gamma_c D)$	T	T	N	1324	T_{11}^{3pt}	$\tilde{T}_{11}^{3pt}$
	T_{12}^{3pt}			N	T	N	1324	T_{11}^{3pt}	$\tilde{T}_{11}^{3pt}$

5 Data layout in *meson-baryon-factorized*

5.1 Nomenclature

sp_gamma_list

- $\mathbb{1}$
- γ_5
- γ_t
- $\gamma_t \gamma_5$

sequential_source_gamma_list

- this is Γ_{i_2} , the gamma structure at sequential source vertex; used to be γ_5 only, now also includes $\gamma_t \gamma_5$
- γ_5
- $\gamma_t \gamma_5$

va_current_gamma_list

- $\gamma_j, j = x, y, z$
- $\gamma_j \gamma_5$

nucleon_gamma_list

- C
- $C \gamma_5$
- $C \gamma_5 \gamma_t$
- $C \gamma_t$

delta_gamma_list

- $C \gamma_j, j = x, y, z$
- $C \gamma_j \gamma_t$
- $C \gamma_j \gamma_t \gamma_5$

delta_gamma_list Union of nucleon_gamma_list and delta_gamma_list

Sources and Propagators

- *fl* forward-light
- *sll* sequential light-after-light
- *phil* stochastic timeslice propagator for light quark
- *phil* oet-stochastic timeslice propagator for light quark

The gamma lists are collected in file `gamma_lists.lua` as tables with entries

$$\{tag="g5", gamma=\{ 15, 1, 1\}\},$$

- *tag* is the name of the product of gamma matrix combinations; depending on context we use mnemonic names like “Cg5”
- *gamma* has three entries
 1. gamma id in binary counting; accessed by QUA’s `gamma{n=id}`
 2. the gamma sign; only relevant for products made of the four gamma matrices $\gamma_{x,y,z,t}$; e.g. $\gamma_t \gamma_x$ is `-gamma{n=9}`, since binary counting means $\gamma_x^{b_0} \gamma_y^{b_1} \gamma_z^{b_2} \gamma_t^{b_3}$ and $b_0 b_1 b_2 b_3$ is the binary word identifier
 3. the adjoint sign; using $\Gamma^\dagger = \sigma_{\Gamma^\dagger} \Gamma$ for all 16 gamma matrices

Momentum lists they are in file `momentum_lists.lua`; we keep

- `sequential_source_momentum_list =  $\vec{p}_{i_2}$`
- `sink_momentum_list =  $\vec{p}_{f_1}, \vec{p}_{f_2}$`

Note: for our purpose here these lists are chosen identical

Placeholders

- `pi2xPPpi2yPPpi2zPP`; PP is at 2 digits integer plus potential sign; e.g. PP = {00, 01, -01}
- `gfGG`, GG is 2 digits integer, $0 \leq GG \leq 15$
- `giGG` same as gf
- `tTTxXXyYYzZZ`, TT, XX, YY, ZZ at least 2 digits integer, e.g. TT = {00, 01, 27, 123}
- `PXd_PYd_PZd`, d is at least single digit integer plus potential sign, e.g. d = {0, 1, -1}
- `cCC`, CC is at least single digit integer, $0 \leq CC < \text{length}(vX)$
- `sampleRR`, RR at least 2-digit integer, e.g. RR = {00, 01, 11}
- `tsrcTT`, TT is at least 2-digit integer, e.g. TT = {00, 01, 23, 123}
- `dphiD`, D is single digit integer, $0 \leq D < 4$

Comments on data keys to follow

- **γ_5 -hermiticity** γ_5 -factors are **already included** in the key's data; this concerns
 1. the oet contractions (the $\phi_\alpha^\dagger$ is actually $(\gamma_5)_{\alpha\beta} \phi_\beta^\dagger \gamma_5$);
 2. contractions made of $\mathcal{V}_3$ with *stochastic propagator* and forward propagator;
 3. contractions made of $\mathcal{V}_2, \mathcal{V}_4$ with *stochastic source* and forward / sequential propagator;

in the latter two cases the key prefix carries a “g5.” preceding the “phi” and “xi” to make it clear

- **all** Fourier transforms **at source, current insertion and sink** are done uniformly with $\exp(i\vec{p}_A \vec{x})$ for $A = i_2, f_1, f_2, c$; thus fixing total momentum $\vec{P}$ and momentum conservation mean

$$\text{2pt } \vec{p}_{f_1} + \vec{p}_{f_2} = \vec{P} = -\vec{p}_{i_1} - \vec{p}_{i_2} \text{ for 2-point functions}$$

$$\text{3pt } \vec{p}_f + \vec{p}_c = \vec{P} = -\vec{p}_{i_1} - \vec{p}_{i_2} \text{ for 3-point functions}$$

- we keep an oet sample counter; but at this point we only do one stochastic oet source (with spin dilution)
- single-hadron 2-point functions (`contract_2pt_light`) and oet contractions (`contract_oet_m_m_2pt_light`, `contract_oet_v{2,3,4}_light`) are done per coherent source location and carry the coherent source location string in the data key;
all other contractions are done and written per base source location and carry the base source location label in the data key;
for the time being we do not use coherent sources, or equivalently we set number of coh. sources eq. 1; which means base source = coherent source in our counting

5.2 `contract_2pt_light`

`contract_2pt_light.NNNN.tbaseTT.aff`

Keys:

1. `/fl-fl/tTTxXXyYYzZZ/FF/gfGG_giGG/PXd_PYd_PZd`
 - $FF \in \{m1, t1, t2\}$
 - gf, gi from
 `m1 sp_gamma_list`
 `t1,t2 nucleon_delta_gamma_list`
 - `tTTxXXyYYzZZ` gives the *coherent source location*
2. `/fl-gc-fl-gi/tTTxXXyYYzZZ/FF/gfGG_giGG/PXd_PYd_PZd`
 - $FF \in \{m1\}$
 - gf from `va_current_gamma_list`
 - gi from `sp_gamma_list`
 - `tTTxXXyYYzZZ` gives the *coherent source location*

- contraction with source smearing, but *without sink smearing*
3. /U-V-W/pi2xPPpi2yPPpi2zPP/tTTxXXyYYzZZ/FF/gfGG_giGG/cCC/PXd.PYd.PZd
 - U-V-W from
 - fl-fl-sll
 - fl-sll-fl
 - sll-fl-fl
 - FF $\in \{t1, t2\}$
 - gf, gi from nucleon_delta_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*

5.3 contract_v3_xi_light

contract_v3_xi_light.NNNN.tbaseTT.aff

Keys:

1. /xil-gf-fl/tTTxXXyYYzZZ/sampleRR/v3/gfGG/cCC/PXd.PYd.PZd
 - gf from sp_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*
2. /xil-gc-fl/tTTxXXyYYzZZ/sampleRR/v3/gfGG/cCC/PXd.PYd.PZd
 - gf from va_current_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*
3. /xil-gf-sll/pi2xPPpi2yPPpi2zPP/tTTxXXyYYzZZ/sampleRR/v3/gfGG/cCC/PXd.PYd.PZd
 - gf from sp_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*
4. /xil-gc-sll/pi2xPPpi2yPPpi2zPP/tTTxXXyYYzZZ/sampleRR/v3/gfGG/cCC/PXd.PYd.PZd
 - gf from va_current_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*

5.4 contract_v2_xi_light

contract_v2_xi_light.NNNN.tbaseTT.aff

Keys:

1. /g5.xil-gf-fl-fl/tTTxXXyYYzZZ/sampleRR/v2/gfGG/cCC/PXd.PYd.PZd
 - gf from nucleon_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*
2. /g5.xil-gf-U-V/pi2xPPpi2yPPpi2zPP/tTTxXXyYYzZZ/sampleRR/v2/gfGG/cCC/PXd.PYd.PZd
 - U-V is from

- fl-sll
- sll-fl
- gf from nucleon_gamma_list
- tTTxXXyYYzZZ gives the *base source location*

5.5 contract_v4_xi_light

contract_v4_xi_light.NNNN.tbaseTT.aff

Keys:

1. /g5.xil-gf-fl-fl/tTTxXXyYYzZZ/sampleRR/v4/gfGG/cCC/PXd_PYd_PZd
 - gf from nucleon_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*
2. /g5.xil-gf-U-V/pi2xPPpi2yPPpi2zPP/tTTxXXyYYzZZ/sampleRR/v4/gfGG/cCC/PXd_PYd_PZd
 - U-V is from
 - fl-sll
 - sll-fl
 - gf from nucleon_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*

5.6 contract_v3_phi_light

contract_v3_phi_light.NNNN.tbaseTT.RRRRR.aff

Keys:

1. /g5.phil-gf-fl-fl/tTTxXXyYYzZZ/sampleRR/v3/gfGG/cCC/PXd_PYd_PZd
 - gf is from sp_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*

5.7 contract_v2_phi_light

contract_v2_phi_light.NNNN.tbaseTT.RRRRR.aff

Keys:

1. /phil-gf-fl-fl/tTTxXXyYYzZZ/sampleRR/v2/gfGG/cCC/PXd_PYd_PZd
 - gf is from nucleon_gamma_list,
 - tTTxXXyYYzZZ gives the *base source location*

5.8 contract_v4_phi_light

contract_v4_phi_light.NNNN.tbaseTT.RRRRR.aff

Keys:

1. /phil-gf-fl-fl/tTTxXXyYYzZZ/sampleRR/v4/gfGG/cCC/PXd_PYd_PZd
 - gf is from nucleon_gamma_list,
 - tTTxXXyYYzZZ gives the *base source location*

5.9 contract_v3_phi_light.dt

contract_v3_phi_light.NNNN.tbaseTT.dtDD.RRRRR.aff

Keys:

1. /g5.phil-gc-fl-fl/tTTxXXyYYzZZ/sampleRR/v3/gfGG/cCC/PXd_PYd_PZd
 - gf from va_current_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*
 - contractions with source smearing, but *without sink smearing*

5.10 contract_v2_phi_light.dt

contract_v2_phi_light.NNNN.tbaseTT.dtDD.RRRRR.aff

Keys:

1. /phil-gf-fl-fl/tTTxXXyYYzZZ/sampleRR/v2/gfGG/cCC/PXd_PYd_PZd
 - gf from nucleon_delta_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*

5.11 contract_v4_phi_light.dt

contract_v4_phi_light.NNNN.tbaseTT.dtDD.RRRRR.aff

Keys:

1. /phil-gf-fl-fl/tTTxXXyYYzZZ/sampleRR/v4/gfGG/cCC/PXd_PYd_PZd
 - gf from nucleon_delta_gamma_list
 - tTTxXXyYYzZZ gives the *base source location*

5.12 contract_oet_m_m_2pt_light

contract_oet_m_m_2pt_light.NNNN.tbaseTT.RRRRR.aff

Keys:

1. /fl-fl/pi2xPPpi2yPPpi2zPP/tsrcTT/m1/gfGG_giGG/PXd_PYd_PZd
 - gf is from sp_gamma_list
 - gi is from sp_gamma_list
 - tsrcTT gives the *coherent source timeslice*
2. /fl-gc-fl-gi/pi2xPPpi2yPPpi2zPP/tsrcTT/m1/gfGG_giGG/PXd_PYd_PZd
 - gf is from va_current_gamma_list
 - gi is from sp_gamma_list
 - tsrcTT gives the *coherent source timeslice*
 - contractions with source smearing, but *without sink smearing*

5.13 contract_oet_v3_light

contract_oet_v3_light.NNNN.tbaseTT.RRRRR.aff

Keys:

1. /phil-gf-fl/pi2xPPpi2yPPpi2zPP/tsrcTT/dphiD/v3/gfGG/cCC/PXd_PYd_PZd
 - gf is from sp_gamma_list
 - tsrcTT gives the *coherent source timeslice*
 - dphiD gives the oet-source spin component
2. /phil-gc-fl/pi2xPPpi2yPPpi2zPP/tsrcTT/dphiD/v3/gfGG/cCC/PXd_PYd_PZd
 - gf is from va_current_gamma_list
 - tsrcTT gives the *coherent source timeslice*
 - dphiD gives the oet-source spin component

5.14 contract_oet_v2_light

contract_oet_v2_light.NNNN.tbaseTT.RRRRR.aff

Keys:

1. /phil-gf-fl-fl/pi2xPPpi2yPPpi2zPP/tsrcTT/dphiD/v2/gfGG/cCC/PXd_PYd_PZd
 - gf is from nucleon_gamma_list,
 - tsrcTT gives the *coherent source timeslice*
 - dphiD gives the oet-source spin component
 - pi2xPPpi2yPPpi2zPP is pi2x00pi2y00pi2z00, since we only need to measure for $\vec{p}_{i_2} = 0$

5.15 contract_oet_v4_light

contract_oet_v4_light.NNNN.tbaseTT.RRRRR.aff

Keys:

1. /phil-gf-fl-fl/pi2xPPpi2yPPpi2zPP/tsrcTT/dphiD/v4/gfGG/cCC/PXd_PYd_PZd
 - gf is from nucleon_gamma_list,
 - tsrcTT gives the *coherent source timeslice*
 - dphiD gives the oet-source spin component
 - pi2xPPpi2yPPpi2zPP is pi2x00pi2y00pi2z00, since we only need to measure for $\vec{p}_{i_2} = 0$

6 Naming of Diagram types

For all B , W , Z -type ι diagrams considered here we join two factors

1. 1 factor of $\mathcal{V}_3$ type with one spinor index;
2. 1 factor of $\mathcal{V}_2$ or $\mathcal{V}_4$ type with 3 spinor indices.

This always takes the form

$$\mathcal{D}(\pi, \text{RE}, \text{OP}_{i_1}, \text{OP}_{f_1})_{\sigma_3, \sigma_4} = \sum_{\sigma_1, \sigma_2} \left(\Gamma_{i_1}^{\text{OP}_{i_1}} \right)_{\sigma_1 \sigma_2} \mathcal{V}_{\text{RE}} \left(\Gamma_{f_1}^{\text{OP}_{f_1}} \right)_{\sigma_{\pi(1)} \sigma_{\pi(2)} \sigma_{\pi(3)}} \mathcal{V}_{3\sigma_{\pi(4)}} \quad (197)$$

Here

- π is a permutation of 1, 2, 3, 4
- RE is the reduction type 2 or 4
- $\text{OP}_{i_1, f_1} = N, T$ without transposition / with transposition;
the transposition OP_{i_1} is determined by the permutation π

The meaningful permutations to be distinguished are

no. of π	$\sigma_{\pi(1)}$	$\sigma_{\pi(2)}$	$\sigma_{\pi(3)}$	$\sigma_{\pi(4)}$
1	σ_1	σ_3	σ_4	σ_2
2	σ_3	σ_1	σ_4	σ_2
3	σ_3	σ_4	σ_1	σ_2
4	σ_1	σ_2	σ_3	σ_4
5	σ_1	σ_3	σ_2	σ_4
6	σ_3	σ_1	σ_2	σ_4

The remaining permutations to reach the full 24 (= 6, which we have, times 2 times 2 =) are obtained by

1. exchange $\sigma_1 \leftrightarrow \sigma_2$ equivalent to $\text{OP}_{i_1} N \leftrightarrow T$;
we don't include these permutations, since we usually always have $\Gamma_{i_1}^t = \pm \Gamma_{i_1}$;
2. exchange $\sigma_3 \leftrightarrow \sigma_4$ equivalent to transposition of final diagram

and need not be calculated.

Note: In our standard calculation we will **not** do recombination for different OP_{i_1, f_1} s; we always assume $\Gamma_{i_1, f_1}^t = \pm \Gamma_{i_1, f_1}$, which is true for all 16 gamma matrices built from $\gamma_{x, y, z, t}$.

Diagram name We want to encode in the nomenclature

- what we call the type or “topology”, i.e. $B, W, Z, T, D, \dots$
- the flavor structure
this is not really of concern for the Wilson-type calculation in the light quark sector ($\pi - N$),
but better to have it
- within the topology and flavor structure the type of spin contraction;
this is given by the permutation and by $\mathcal{V}_2, \mathcal{V}_4$

- (all the rest coming from the 2/3/4 vertices)

We take the easy way and just join the information from the two factors. So we choose the following way:

$$\text{Diagram name} = RN/PX - G - P/RN/PX - G - P - P/p\sigma_{\pi(0)}\sigma_{\pi(1)}\sigma_{\pi(2)}\sigma_{\pi(3)} \quad (198)$$

Here we have

- R is the reduction type $R = v, t, \dots$
- N is a reduction type $N = 2, 3, 4, \dots$
- P means seq⁽ⁿ⁾uential propagator
 - u, d - forward up, down
 - sll - sequential light-after-light
 - sud - sequential up-after-down
 - that's all we need for now
- PX means stochastic source or propagator
 - $phil$ - light stochastic propagator
 - $g5.phil$ - light stochastic propagator multiplied by γ_5 ; in case γ_5 -hermiticity was used
 - $phiu, phis$ - up, strange tochastic propagator
 - xil - light stochastic source
 - $g5.xil$ - light stochastic source multiplied by γ_5 ; in case γ_5 -hermiticity was used
 - xiu, xis - up, strange stochastic source (in case they are differently smeared or so)
 - that's all we need for now
- G is a string for type of vertex, whether with
 - smeared propagtor ends joining - gf
 - non-smeared propagator ends joining - gc
- $p\sigma_{\pi(0)}\sigma_{\pi(1)}\sigma_{\pi(2)}\sigma_{\pi(3)}$, i.e. p followed by four numbers 0,1,2,3 in some order denotes the permutation used to build the diagram

Examples In general, the diagram name can be read off directly from its composition of reduction operations.

- B -type $v3/xil - gf - sll/v2/phil - gf - u - u/p0123$
2-point function box-type diagram, built from $\mathcal{V}_3$ including sequential and $\mathcal{V}_2$
- W -type $v3/g5.phil - gc - u/v4/g5.xil - gf - u - sll/p2301$
3-point function W -type diagram, built from $\mathcal{V}_3$ including forward and current insertion vertex, and from $\mathcal{V}_4$
- direct diagram $/m1/fl - fl/t2/fl - fl/p0000$
- and so on

6.1 Code

This scheme is used by
cvc code,
branch piN2piN-matrix-elements,
program meson_baryon_factorized_diagrams_3pt.cpp.